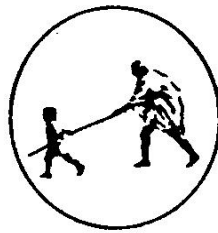


An Infosys Virtual Internship Report on
Edu2Job Predicting Job Roles from Educational
Background

BY
BUDDHEWAD ADVITH DEEPAK

Under the Guidance
of
Mr. Shivprasad Titare

(Department of Computer Science and Engineering)



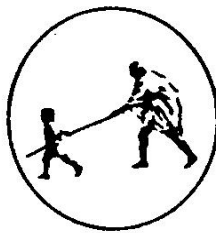
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)

Academic Year 2025-26

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Submitted to
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For fulfillment of the requirement for the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING
By
BUDDHEWAD ADVITH DEEPAK

Under the Guidance
of
Mr. Shivprasad Titare

(DEPARTMENT OF COMPUTER SCIENCE ENGINEERING)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)

Academic Year 2025-26

Certificate



This is to certify that the project entitled

“Edu2Job Predicting Job Roles from Educational Background”

*being submitted by **Mr. Buddhewad Advith Deepak** to the Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by him under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

Mr. Shivprasad Titare

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H.O.D

Computer Science & Engineering

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Director

MGM's College of Engg., Nanded

OFFER LETTER



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1 message

Infosys Springboard-Support <Springboard-support@infosys.com>

Wed, 11 Feb, 2026 at 16:51



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ACKNOWLEDGEMENT

I would like to express my deepest gratitude to my project guide, **Mr. Shivprasad Titare**, for his invaluable support, guidance, and encouragement throughout this project. His profound knowledge and expertise have been instrumental in the successful completion of this work. His patience and willingness to assist me at every step have greatly enriched my learning experience. His constructive feedback and insightful suggestions have not only helped me overcome challenges but also motivated me to strive for excellence.

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Last but not least, I am also thankful to all those who helped, directly or indirectly, develop this project and complete it successfully.

With Deep Reverence,

Buddhewad Advith Deepak

[B.Tech CSE-B]

ABSTRACT

The Edu2Job – AI Career Intelligence System is a web-based career guidance and employability enhancement platform developed to assist students and job seekers in making informed career decisions. Traditional career guidance methods often rely on manual counseling, generic recommendations, and limited industry insights, which may lead to confusion in career selection and inadequate preparation for future job opportunities. The developed system addresses these challenges by providing an intelligent and personalized platform that combines Machine Learning, Artificial Intelligence, resume analysis, skill-gap identification, and career roadmap generation.

The application was developed using modern full-stack technologies including React.js, FastAPI, MongoDB, Gradient Boosting Classifier, Gemini AI, and REST APIs. The frontend provides an interactive and user-friendly interface for students and administrators, while the backend handles business logic, Machine Learning predictions, resume processing, and API communication. MongoDB is utilized for secure storage and management of user profiles, career prediction data, resume analysis results, and learning recommendations.

The system provides several important functionalities such as secure user authentication, AI-based career prediction, resume analysis, skill-gap assessment, personalized career roadmap generation, course recommendations, and job market insights.

The platform also generates structured career roadmaps that guide users through learning paths, certifications, projects, and professional development activities required to achieve their desired career goals. Administrators can monitor user activities, prediction statistics, skill trends, and platform analytics through dedicated dashboards. By integrating Artificial Intelligence and Machine Learning technologies, the Edu2Job system provides data-driven career guidance that helps bridge the gap between academic education and industry requirements, ultimately improving employability and career readiness.

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INTRODUCTION

In today's fast-changing digital world, students often have trouble choosing the right career path based on their academic background, technical skills, certifications, and interests. Traditional career guidance relies mainly on manual counseling, aptitude tests, or fixed recommendations. These methods tend to be slow, less personalized, and struggle to keep up with changing industry demands. As the technology industry grows, we need smart systems that can help students find suitable job roles through data-driven analysis.

To solve this problem, we developed the project "Edu2Job – AI Career Intelligence System." It serves as a smart career guidance and job prediction platform powered by Artificial Intelligence (AI) and Machine Learning (ML). The system examines a student's educational background, technical skills, projects, internships, certifications, and resume content to suggest suitable career roles and provide tailored recommendations. The platform helps to connect education with industry needs by showing students which job roles fit their current profile and what improvements they should make to reach their desired careers.

The project combines multiple advanced technologies including Machine Learning, Natural Language Processing (NLP), Generative AI, FastAPI, React, and MongoDB to create a complete end-to-end career intelligence platform. The Machine Learning model predicts suitable job roles based on structured academic and skill-related inputs, while the AI-powered modules provide resume analysis, skill-gap identification, and personalized career improvement suggestions. The system also includes an Admin Dashboard for monitoring user activity, prediction statistics, and system analytics.

1.1 Overview of the Project

Edu2Job – AI Career Intelligence System is an intelligent web-based platform developed to help students and fresh graduates identify suitable career opportunities based on their educational background, technical skills, projects, internships,

certifications, and resume details. In the current competitive industry environment, many students face confusion while selecting career paths because traditional guidance systems are mostly manual, generalized, and unable to provide personalized recommendations according to individual profiles. This creates a significant gap between academic learning and industry requirements.

The proposed system aims to bridge this gap by using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) technologies to provide smart career guidance and job role prediction. The system collects various user details such as CGPA, specialization, technical skills, certifications, internships, academic performance, and resume content, and processes them using a Machine Learning prediction engine. Based on this analysis, the platform predicts suitable job roles such as Frontend Developer, Backend Developer, Full Stack Developer, Data Scientist, Machine Learning Engineer, Cybersecurity Analyst, and Software Engineer.

The project combines modern technologies including React.js for frontend development, FastAPI for backend services, MongoDB for database management, and Random Forest Classifier for Machine Learning predictions. Additionally, Gemini AI is integrated into the system to perform advanced resume analysis, skill-gap identification, and personalized career recommendations. The platform also provides suggestions for certifications, learning resources, projects, and technologies required to improve career readiness.

The system architecture is designed as a complete end-to-end career intelligence platform consisting of multiple modules such as Job Prediction, Resume Analyzer, Skill Gap Analysis, AI Career Guidance, and Admin Dashboard. The Admin Dashboard allows administrators to monitor user activity, role distribution analytics, prediction statistics, and system performance. The frontend communicates with the backend using REST APIs, while the backend processes data, interacts with Machine Learning models, and generates intelligent responses.

1.2 Problem Statement

In the modern education system, students and fresh graduates often face significant difficulties while selecting suitable career paths and preparing themselves according to industry requirements. Although students complete academic programs successfully, many of them remain uncertain about which job roles best match their

educational background, technical skills, certifications, project experience, and personal interests. This confusion creates a major gap between academic learning and professional career development. Students frequently apply for jobs without understanding whether their profiles align with the expectations of the industry, which leads to poor placement outcomes, lack of confidence, and career dissatisfaction.

Traditional career guidance methods mainly depend on manual counseling, aptitude tests, and generalized recommendations. These methods are usually time-consuming, expensive, and unable to provide personalized suggestions for every student. Career counselors may not always analyze complete academic records, technical skills, project quality, internship experience, certifications, and resume content in detail. As a result, many students receive generic advice that does not accurately represent their strengths, weaknesses, or career potential. Another major issue is the rapid growth of technology and continuously changing industry demands. Modern industries require practical knowledge and skills in areas such as Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, Full Stack Development, Cybersecurity, and DevOps. However, many students remain unaware of current market requirements and trending technologies. Academic institutions mainly focus on theoretical education, while industries expect practical implementation skills, real-world project experience, and updated technical knowledge. This creates a significant skill gap between students and industry expectations.

Resume preparation and screening is another major challenge for students and recruiters. Many students create resumes without understanding professional standards, ATS requirements, or proper technical presentation. Due to missing technical keywords, weak project descriptions, improper formatting, and lack of clarity, students often fail to receive interview opportunities even if they possess good technical abilities.

Another important challenge is the absence of integrated systems that combine job prediction, resume analysis, skill-gap identification, AI career guidance, and analytics into a single platform. Most currently available systems focus only on one or two functionalities, forcing users to depend on multiple separate platforms for career preparation. This creates inconvenience, inconsistency, and incomplete analysis of the student profile.

To overcome these challenges, there is a strong need for an intelligent AI-powered platform capable of analyzing complete student profiles, predicting suitable job roles, identifying missing skills, performing detailed resume analysis, and generating personalized career recommendations. The proposed Edu2Job – AI Career Intelligence System aims to address these problems by integrating Machine Learning, Natural Language Processing, and Generative AI technologies into a unified platform. The system intelligently analyzes educational and technical information, predicts career opportunities, identifies skill gaps, and provides personalized guidance to help students become industry-ready and make informed career decisions.

1.3 Objectives of the Project

To systematically address the challenges identified in career guidance, employability enhancement, and job prediction, the Edu2Job – AI Career Intelligence System was designed with the following objectives:

- **Career Role Prediction:** Develop a Machine Learning-based prediction system capable of analyzing academic records, technical skills, certifications, projects, internships, and career interests to recommend the most suitable job roles for students.
- **Resume Analysis and Evaluation:** Implement an intelligent resume analysis module that extracts important information from resumes, evaluates profile strength, identifies missing skills, and provides recommendations for resume improvement using Natural Language Processing and Generative AI techniques.
- **Skill Gap Identification:** Compare user competencies with industry requirements and identify missing technical and professional skills required for target career roles, helping users understand areas that need improvement.
- **Personalized Career Roadmap Generation:** Generate structured learning paths that guide users through certifications, projects, internships, technologies, and practical activities required to achieve their desired career goals. The roadmap also helps users track their progress and prioritize learning objectives according to industry demands and career aspirations.

- **Course Recommendation System:** Recommend relevant courses, learning resources, and certification programs based on career predictions and skill-gap analysis results to support continuous skill development.
- **User-Friendly Web Application:** Develop a responsive and interactive web platform using React.js and FastAPI that provides seamless navigation, secure authentication, and efficient communication between system modules.
- **Scalable and Secure Architecture:** Build a reliable, scalable, and secure system capable of handling multiple users while ensuring data privacy, efficient processing, and future extensibility.
- **Admin Dashboard and Analytics:** Design an administrative dashboard that enables monitoring of user activities, prediction statistics, skill trends, platform usage, and overall system performance through visual analytics and reports.

The main objective of the Edu2Job project is to develop an intelligent and user-friendly career guidance platform that helps students identify suitable job roles based on their academic background, technical skills, certifications, internships, projects, and resume details. The system is designed to reduce the confusion students face while selecting career paths and preparing for placements by providing accurate and personalized recommendations using Artificial Intelligence and Machine Learning technologies.

Another important objective is to perform intelligent resume analysis using Natural Language Processing and Generative AI. The platform should be able to analyze resumes, extract important information, identify missing skills, and provide suggestions for improving resume quality according to current industry standards. The project also focuses on identifying skill gaps between the student's current profile and the skills required in the industry. Many students are unaware of the technologies and practical skills expected for different job roles. Therefore, the system aims to provide personalized recommendations regarding certifications, technologies, projects, and learning resources required for career improvement.

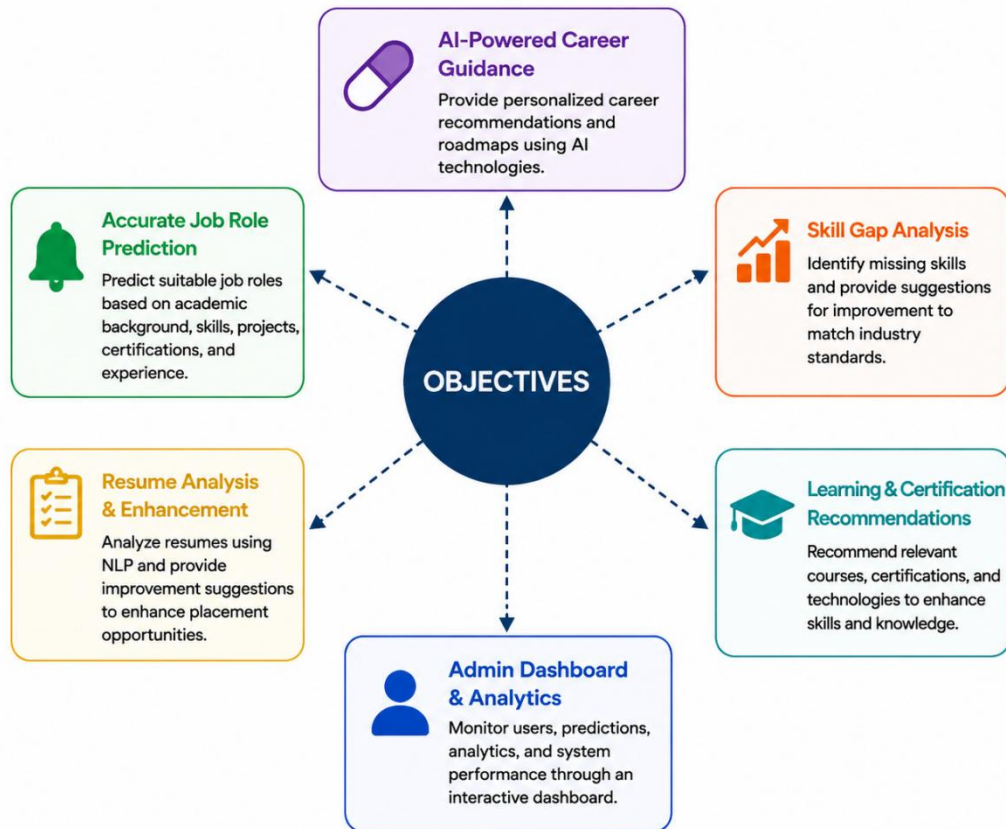


Fig. 1.1: Objectives of the Project

The above figure illustrates the complete workflow of the Edu2Job system. The process begins with user input, where students provide educational details, technical skills, projects, internships, certifications, and resume information. The system then performs data preprocessing and validation before sending the processed data to the Machine Learning prediction engine. The Random Forest model predicts suitable job roles, while the AI modules perform resume analysis and skill-gap identification. Finally, the system generates personalized career recommendations, learning roadmaps, and improvement suggestions for the user.

The project also aims to develop a modern and responsive web application using React.js and FastAPI. The platform should provide a smooth user experience, secure authentication, efficient backend processing, and real-time API communication between different modules. The system should be scalable, reliable, and easy to use for students as well as administrators. Another objective of the project is to implement an Admin Dashboard for monitoring system activities and user analytics.

1.4 Scope of the Project

The scope of the Edu2Job project is centered around developing an intelligent and practical career guidance platform that helps students and fresh graduates make better career decisions using Artificial Intelligence and Machine Learning technologies. The system is designed to analyze educational and technical information and provide personalized career recommendations based on the user's profile. The project mainly focuses on solving the gap between academic learning and industry requirements by helping students understand which job roles are most suitable for them and what skills they need to improve.

The platform supports Machine Learning-based job role prediction where users can enter details such as educational qualifications, specialization, technical skills, project experience, certifications, internships, and subject knowledge. Based on these inputs, the system predicts suitable career roles in the software and technology domain.

Another important area within the scope of the project is AI-powered resume analysis. The system allows users to upload resumes, which are then analyzed using Natural Language Processing and Generative AI techniques. The platform extracts important information from resumes, identifies technical skills and provides suggestions for improvement. The project also includes Skill Gap Analysis functionality. This feature compares the student's current skills with the skills required for a target job role and identifies missing areas. Based on the analysis, the system recommends suitable certifications, technologies, projects, and learning resources that can help students improve their profiles and increase placement opportunities. This makes the platform more practical and beneficial for career preparation.

The project also includes a complete Admin Dashboard for system monitoring and analytics. The dashboard helps administrators manage users, monitor prediction activities, analyze role distribution statistics, and track overall platform performance through charts and reports. This feature makes the system useful not only for students but also for educational institutions and placement departments. The scope of the project is not limited to academic implementation only. The system architecture is scalable and can be expanded in the future to support additional domains such as

healthcare, finance, management, marketing, and non-technical career fields. New Machine Learning models, advanced AI features, real-time recruitment integration, and cloud deployment can also be added in future versions.

Overall, the Edu2Job platform provides a strong foundation for intelligent career guidance and placement preparation. The project demonstrates how modern AI technologies can be used effectively to create smart educational and career support systems that are practical, scalable, and useful in real-world applications.

1.5 Existing Technologies and Solutions

In recent years, many career guidance and recruitment platforms have been developed to help students and professionals identify suitable job opportunities and improve their career preparation. These systems mainly focus on resume screening, aptitude testing, job recommendations, and online recruitment services. Although these platforms provide useful functionalities, most of them still have limitations in terms of personalization, intelligent analysis, and complete career guidance.

Traditional career guidance systems mainly depend on manual counseling and aptitude tests. In these systems, career experts interact with students and provide suggestions based on academic performance, interests, or personality assessments. While these methods are helpful, they are often time-consuming and unable to provide personalized analysis for every student. Manual guidance also depends heavily on the knowledge and experience of counselors, which may lead to generalized recommendations instead of accurate career predictions.

Online job portals such as LinkedIn, Naukri, and Indeed provide job recommendations based on resumes, user profiles, and search activities. These platforms are mainly designed for recruitment and job searching rather than detailed career guidance. Although they help users apply for jobs, they do not deeply analyze academic performance, technical skills, projects, internships, or certifications while generating recommendations. As a result, students often receive suggestions that may not fully match their profiles or career goals.

Several resume analyzer tools are also available in the market. They help users improve resume structure and identify missing keywords according to job descriptions. However, most existing resume analyzers do not provide complete profile evaluation,

career alignment analysis, or personalized learning recommendations. They generally focus only on resume optimization rather than long-term career development.

With the advancement of Artificial Intelligence and Machine Learning technologies, modern recommendation systems have become more intelligent and data-driven. Machine Learning algorithms are capable of analyzing large datasets and identifying patterns to generate predictions and recommendations. These technologies are widely used in recommendation systems, recruitment platforms, e-learning applications, and predictive analytics systems. Natural Language Processing techniques further improve these systems by enabling text extraction, semantic analysis, resume parsing, and intelligent understanding of user input.

These AI models can generate personalized responses, career roadmaps, interview preparation strategies, and learning suggestions according to user requirements. Generative AI improves user interaction and provides more human-like guidance compared to traditional static systems. Despite these technological advancements, most existing solutions focus only on specific functionalities such as job searching, resume checking, or online learning recommendations. Very few systems combine Machine Learning-based job prediction, AI-powered resume analysis, skill-gap identification, personalized career guidance, and analytics into one integrated platform. Students often need to use multiple separate applications for career preparation, which creates inconvenience and incomplete analysis.

- **Advantages of Existing Technologies and Solutions:** Existing career guidance and recruitment platforms provide several useful features that help students and professionals during their career development process. Online job portals offer access to thousands of job opportunities across different industries and locations. Resume analyzer tools help users optimize their resumes according to industry standards and improve keyword matching for recruitment systems. Machine Learning-based recommendation systems can analyze large amounts of data and generate personalized suggestions efficiently. Natural Language Processing technologies enable automated resume parsing, skill extraction, and profile evaluation. Generative AI technologies further enhance user experience by providing interactive guidance, learning recommendations, and career-

related insights. These technologies have significantly improved the accessibility and efficiency of career planning and job searching processes.

- **Disadvantages of Existing Technologies and Solutions:** Despite their benefits, existing solutions have several limitations. Most career guidance platforms focus only on a single functionality such as job searching, resume evaluation, or online learning recommendations. Traditional career counseling methods are time-consuming and often provide generalized suggestions rather than personalized guidance. Job portals mainly recommend jobs based on profile matching and do not perform detailed analysis of academic performance, technical skills, projects, certifications, and internships. Resume analyzer tools generally focus on formatting and keyword optimization without evaluating overall career readiness. Many systems lack skill-gap identification and do not provide structured roadmaps for career improvement. Additionally, users often need to use multiple separate platforms for career prediction, resume analysis, learning recommendations, and skill development, resulting in fragmented guidance and an incomplete understanding of their career progression.

The proposed Edu2Job system aims to overcome these limitations by integrating multiple intelligent technologies into a single unified platform. The system combines Machine Learning for job prediction, Natural Language Processing for resume analysis, Generative AI for personalized career guidance, and modern web technologies for seamless user interaction. Unlike existing systems, Edu2Job not only predicts suitable job roles but also explains skill gaps, recommends learning resources, and helps students improve their placement readiness through intelligent analysis and personalized recommendations.

1.6 Features of the System

The Edu2Job system is designed as an intelligent and interactive career guidance platform that combines Artificial Intelligence, Machine Learning, and modern web technologies to help students make better career decisions. The system provides multiple advanced features that assist users in understanding their career opportunities, improving their technical profiles, and preparing for placements more effectively.

One of the main features of the system is Machine Learning-based Job Role Prediction. The platform analyzes the user's educational qualifications, technical skills, certifications, internships, projects, and academic performance to predict suitable career roles. The prediction engine uses a trained Machine Learning model to identify job roles that best match the user's profile. This helps students understand which career paths align with their current skills and knowledge.

Another important feature of the system is AI-powered Resume Analysis. Users can upload their resumes, and the system automatically analyzes the content using Natural Language Processing and Generative AI techniques. The platform extracts technical skills, project details, certifications, and important keywords from the resume and evaluates its quality according to industry standards. It also provides suggestions to improve resume structure, content, and technical presentation.

The Skill Gap Analysis feature helps students identify the difference between their current abilities and the skills required for their target job roles. The system compares the user's profile with industry expectations and highlights missing technologies, tools, or concepts. Based on this analysis, the platform recommends certifications, courses, projects, and learning resources that can help students improve their technical knowledge and placement readiness.

The system also includes AI-based Career Guidance using Generative AI integration. This feature provides personalized career recommendations, interview preparation tips, learning roadmaps, and improvement strategies according to the student's profile. Unlike traditional systems that provide static suggestions, Edu2Job offers more interactive and intelligent guidance tailored to individual users.

Another key feature is the modern and responsive User Interface developed using React.js. The platform is designed to provide a smooth and user-friendly experience across different devices. Users can easily access prediction modules, upload resumes, view analytics, and interact with the system without technical complexity.

The project also includes a secure and scalable Backend System developed using Fantasist. The backend handles API communication, Machine Learning integration, user authentication, data processing, and database management efficiently.

The system supports secure access and smooth interaction between frontend and backend modules.

An important administrative feature of the project is the Admin Dashboard and Analytics module. This dashboard allows administrators to monitor users, analyze prediction statistics, track role distributions, and observe system performance through charts and reports. It helps manage the overall platform effectively and provides valuable insights into user activity and career trends.

The platform also supports Real-Time API Communication between frontend and backend services, ensuring faster response times and seamless data processing. MongoDB is used for storing user information, analytics data, and system records in a flexible and scalable manner. Another valuable feature of the system is Explainable Prediction Results. Instead of simply predicting a job role, the platform also explains why a particular role is recommended by analyzing relevant skills, projects, certifications, and subject knowledge. This improves user understanding and increases trust in the prediction system.

Overall, the Edu2Job platform combines intelligent prediction, resume analysis, skill-gap detection, AI guidance, analytics, and modern web technologies into a single integrated system. These features make the platform highly useful for students, placement departments, and educational institutions aiming to improve career preparation and placement outcomes.

LITERATURE SURVEY

The rapid growth of technology and digital transformation has significantly changed the way career guidance and recruitment processes are carried out. Traditional methods of career counseling mainly relied on manual interaction, aptitude tests, and generalized recommendations. However, with the advancement of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP), modern career guidance systems have become more intelligent, data-driven, and personalized.. However, with the advancement of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP), modern career guidance systems have become more intelligent, data-driven, and personalized.

Out there, picking a path after school feels harder when what you learn clashes with what jobs need. Guidance that once leaned on paper tests now shows its age. Picture advice shaped by slow, one size fits most chats with counselors. Think long waits for feedback that barely fits your story. Modern shifts in work move faster than old methods can keep up. Matching talent to roles stumbles without sharper tools. Personal touch fades when scale takes over. What worked yesterday drags behind today's pace.

Researchers have proposed various intelligent career recommendation systems, resume screening platforms, job matching applications, and Machine Learning-based career prediction models to assist students and job seekers in making informed career decisions. These systems utilize technologies such as Artificial Intelligence, Machine Learning, Natural Language Processing, Data Mining, and Predictive Analytics to analyze user profiles and generate personalized recommendations.

Agarwal and Sharma(2019)proposed a Machine Learning-based career recommendation framework that analyzed educational qualifications, technical skills, and academic performance to suggest suitable career opportunities. Their study demonstrated that supervised learning algorithms can significantly improve career prediction accuracy compared to traditional rule-based systems. However, the system

mainly focused on academic records and did not incorporate resume analysis or industry skill-gap identification.

Kumar et al. (2020) developed an intelligent job recommendation platform using data mining and predictive analytics techniques. The system evaluated user profiles, certifications, and technical competencies to recommend suitable job roles. The research highlighted the importance of personalized recommendations in improving employability. However, the proposed system lacked AI-driven career guidance and personalized learning recommendations.

The research paper titled "AI-Based Career Counseling System Using Machine Learning" explained how Artificial Intelligence can be utilized to automate career counseling processes. The system analyzed academic data, skills, interests, and aptitude scores to generate career suggestions. The study concluded that AI-based career counseling systems can improve decision-making efficiency and provide more personalized career recommendations.

Singh and Verma (2021) proposed a resume screening and candidate assessment platform using Natural Language Processing techniques. Their system automatically extracted skills, educational qualifications, projects, and certifications from resumes to evaluate candidate suitability. The research demonstrated the effectiveness of NLP in reducing manual resume screening efforts. However, the system focused mainly on recruitment processes and did not provide career development guidance for students.

Patel et al. (2022) developed a skill-gap identification system that compared user skills with industry requirements. The platform identified missing competencies and recommended learning resources to improve employability. Their study emphasized the growing importance of continuous skill development in modern job markets. However, the system lacked career prediction capabilities and integrated analytics features.

Recent advancements in Generative AI have introduced new opportunities in career guidance systems. Modern AI models can analyze resumes, identify improvement areas, generate personalized career roadmaps, and recommend learning resources dynamically. These technologies have significantly enhanced the

effectiveness of intelligent career advisory platforms by providing contextual and personalized recommendations.

The literature survey indicates that while existing systems provide functionalities such as career recommendation, resume analysis, job matching, or skill-gap identification individually, very few platforms combine all these capabilities into a single integrated solution. This creates a need for an intelligent platform that can perform career prediction, resume analysis, skill-gap assessment, AI-powered mentoring, and analytics-driven career guidance within a unified environment.

2.1 Study on Career Recommendation Systems

Career recommendation systems are developed to assist students and professionals in identifying suitable career opportunities based on their educational qualifications, technical skills, interests, and professional experience. Traditional career guidance methods often rely on counselors and aptitude tests, which may not always provide personalized recommendations.

Recent research has focused on applying Machine Learning algorithms such as Decision Trees, Random Forests, Support Vector Machines, and Gradient Boosting techniques to improve recommendation accuracy. These systems analyze user profiles and generate data-driven career suggestions based on learned patterns from historical datasets.

Several studies have demonstrated that Machine Learning-based career recommendation systems can significantly improve career planning by providing objective and personalized guidance. However, many existing systems focus primarily on career prediction and do not provide detailed improvement strategies or skill-gap analysis.

The Edu2Job platform extends these capabilities by integrating career prediction with resume analysis, skill-gap identification, AI-powered mentoring, and personalized learning recommendations.

2.2 Study on Resume Analysis Systems

These days, picking through job applications means dealing with piles of resumes. Picture one role attracting countless responses - sorting them by hand takes forever. That slows everything down, uses up energy. So places turn to smart tools that scan

CVs in seconds instead. Machines handle the first look now, spotting who fits without dragging out the process.

Looking at resumes the old way means people must check each one by hand, judging skills, background, education, credentials, along with past projects. Even if it works now and then, reading every detail yourself brings in personal leanings, uneven decisions, plus mental fatigue. Because of that, strong applicants sometimes slip through the cracks without notice. Machines step in here - speeding up reviews while cutting down favoritism and growing room to handle piles of applications fairly.

Out of nowhere, some scientists tried teaching computers to read resumes using tools like named entity spotting and word pattern tracking. Because every job applicant writes differently, these methods help software still catch key details - say, a programming skill buried in odd phrasing or unusual layout. Machines now adjust on their own, pulling out certifications or expertise regardless of how creatively they're described down below. From raw details to future paths - Edu2Job goes beyond scanning resumes. Instead of just reading text, it forecasts fitting job directions based on user data

Researchers have proposed various resume parsing techniques that employ Named Entity Recognition, keyword extraction, semantic analysis, and machine learning algorithms to improve information extraction accuracy. These approaches enable systems to identify domain-specific skills and qualifications even when different terminology or formatting styles are used within resumes.

The Edu2Job platform extends the functionality of traditional resume analyzers by integrating resume evaluation with career prediction, skill-gap identification, and AI-powered recommendations. The system not only extracts and analyzes resume content but also predicts suitable career roles, calculates profile strength scores, identifies missing skills, and generates actionable suggestions for improvement. This transforms resume analysis from a recruitment tool into a comprehensive career development solution.

2.3 Study on Skill Gap Analysis Systems

Right now, many people find themselves short on certain abilities compared to what a new job actually asks for. Tools that spot these differences help point out exactly where someone needs growth. Because tech shifts fast, staying still means falling

behind - learning can't stop once school ends. What used to be enough often isn't anymore when jobs transform. This kind of evaluation plays a quiet but steady part in shaping careers today.

Out there, old-school ways usually mean people rate themselves, get reviewed by others, or have skills mapped by hand. Those things help, yet sometimes opinions creep in - distorting what someone truly knows. Today's tools work differently: pulling real data to line up a person's abilities against what jobs actually need right now. Gaps show up clearly when numbers do the talking instead of guesses..

Most tools that spot skill gaps just list what you're missing - yet they rarely show a clear path to learn it. Some work in isolation, ignoring personal goals or past education along the way. That makes their suggestions hard to use. Their output often misses real-life usefulness because context gets left out.

Starting with where you stand, Edu2Job lines up your current abilities next to what jobs actually demand. Once possible careers pop up via smart algorithms, it checks how well you match each one. The Edu2Job platform addresses these challenges by integrating skill gap analysis directly with career prediction results. After identifying suitable job roles through Machine Learning, the system compares user competencies against industry requirements and generates personalized recommendations. Missing skills, certifications, tools, frameworks, and learning resources are presented in a structured manner, helping users create focused development plans.

Through its intelligent analysis capabilities, the Edu2Job system helps students bridge the gap between academic learning and professional requirements. The platform provides clear guidance on how to improve employability and prepare for desired career opportunities, making skill gap analysis a practical and actionable component of career development.

2.4 Study on Artificial Intelligence in Career Guidance

These days machines help shape job advice just like they do in other fields. Instead of only talking with counselors, people once relied on meetings, school results, test scores to get direction. Even if those ways work sometimes, they struggle under tight schedules, personal opinions, handling too much info at once.

Starting with your background, these tools take in education level, work skills, certificates, likes and dislikes, how you think and act, what jobs are trending, plus hiring stats.

Instead of one-size-fits-all advice, they shape suggestions just for you, shifting as job markets shift. Studies show people make better choices when using smart-guidance apps built on artificial intelligence, reaching many without losing quality. Hidden links inside your details get spotted by learning machines, pointing toward roles that fit now - and later. New leaps in generative models have pushed those tools even further.

Out of nowhere, some groups started rolling out AI tools in schools and job training spots. Because these setups show up, learners and workers get better at spotting where they can grow. A step beyond listing options, the tech sketches paths forward through skill building. When used regularly, preparing for what jobs will ask later becomes less guesswork.

Even with those benefits, most current AI tools for career advice stick to just one thing at a time - like answering questions, checking resumes, or suggesting jobs. It's rare to find a system where forecasting careers, reviewing your background, spotting missing skills, offering mentorship, and guiding study choices all work together. Several organizations have begun implementing AI-powered career development platforms within educational institutions and corporate training environments.

Despite these advantages, many existing AI career guidance solutions focus on individual functionalities such as chatbot support, resume evaluation, or career recommendation. Few platforms provide an integrated ecosystem that combines career prediction, resume analysis, skill-gap assessment, mentoring, and learning recommendations.

The Edu2Job platform utilizes Artificial Intelligence to provide a comprehensive career intelligence solution. Through the integration of Generative AI, Machine Learning, and predictive analytics, the system delivers personalized recommendations, career roadmaps, interview preparation guidance, and profile improvement strategies. This holistic approach enables users to receive continuous support throughout their career development journey.

2.5 Study on Machine Learning-Based Job Prediction Systems

Lately, machines that learn have become a go-to tool for guessing what comes next - especially when sorting jobs and guiding careers. Instead of just listing options, smart tools now study personal backgrounds to suggest fitting roles. Hidden inside these tools? Piles of old records showing how training paths link up with real work results. Spotting repeated trends helps connect degrees, hands-on practice, certificates, abilities, and actual hiring patterns in ways people often miss. What once took weeks of manual matching now happens quietly behind screens.

Most old-school job advice sticks to fixed checklists or human guesswork. Though helpful at first glance, those ways struggle when markets shift or people differ too much. Learning patterns from tons of data lets machines refine their suggestions over time instead. Training and testing boost precision without relying on rigid formulas.

Some studies look at how machines learn to suggest jobs or careers. These methods often pick choices step by step like answering yes-or-no questions. One common way builds tree-shaped paths using personal details to reach a result. It works fast, shows clear reasons, yet can get too tangled if data is messy. When patterns shift slightly, the outcome might change a lot. Built-in logic helps people follow along, though it sometimes fits noise instead of real trends. Predictions split node by node until reaching an answer. If information lacks balance, results wobble even with small changes.

Out of all machine learning methods tested, Gradient Boosting often comes out ahead when measuring precision. When data includes linked elements - things like grades, coding abilities, completed courses, hands-on projects, or past internships - these models handle patterns well. Because they detect subtle links across inputs, outcomes tied to job paths become easier to forecast. What stands out is how finely they untangle overlapping influences within organized information.

Starting with several options, the Edu2Job system tested different models like Decision Trees, SVMs, Naïve Bayes, also Random Forests. Though each had strengths, results showed one stood out when judging precision alongside real-world adaptability. Instead of just high scores, what mattered most was consistent performance across varied datasets. Eventually, Gradient Boosting became the main

choice due to its steady learning pattern and fewer errors over time. Even if slower at times, it handled unseen data better than others did. So now, that method powers the career predictions users receive through the platform.

Internships leave traces in the data, guiding how the system sees practical ability. Certifications add weight, adjusted not by name but by relevance. Skills listed go through resizing so none dominate unfairly. Words about achievements become vectors, letting patterns emerge quietly. Relationships hidden across fields come alive during analysis. Matching students to jobs happens by spotting subtle echoes in their combined traits. Outcomes form not from rules, but from what the full picture suggests. Personal paths unfold based on many small signals gathered along the way..

Furthermore, the Edu2Job platform combines Machine Learning with Natural Language Processing (NLP) and Generative Artificial Intelligence technologies. NLP techniques are used to extract skills, educational qualifications, and experience details from resumes, while Generative AI assists in resume evaluation, skill-gap identification, project recommendations, and personalized career roadmap generation. The integration of these intelligent technologies creates a comprehensive career guidance ecosystem that supports students throughout their career development journey.

2.6 Limitations of Existing Systems

Although existing career guidance platforms provide useful functionalities, several limitations still exist. Most systems focus only on a single aspect of career development such as job recommendation, resume screening, or skill-gap analysis. Users often need multiple platforms to obtain complete career guidance.

Many existing systems lack personalized recommendations and fail to consider the complete profile of the user. Traditional career recommendation tools frequently rely on static rules rather than adaptive Machine Learning models, reducing prediction accuracy and personalization. Several platforms also provide limited analytics and progress tracking capabilities. Students often receive recommendations but do not receive clear guidance on how to improve their profiles or achieve career readiness.

Another significant limitation is the lack of integration between resume analysis, career prediction, skill-gap assessment, AI mentoring, and learning

recommendations. This creates fragmented user experiences and reduces overall effectiveness.

2.7 Need for the Proposed System

The rapid growth of technology and increasing competition in the job market have made career planning a challenging task for students and fresh graduates. Traditional career guidance methods mainly rely on manual counseling, aptitude tests, and generalized recommendations, which often fail to consider a student's complete academic and technical profile. Existing career platforms primarily focus on individual functionalities such as job searching, resume screening, or online learning, requiring users to switch between multiple applications to obtain complete career guidance. As a result, students often struggle to identify suitable career paths and understand industry expectations.

Modern industries expect candidates to possess not only strong academic performance but also practical skills, certifications, project experience, internships, and industry-relevant knowledge. However, many students are unaware of the specific skills and competencies required for different job roles. .

To address these challenges, there is a need for an intelligent and integrated platform capable of providing personalized career recommendations, resume evaluation, skill-gap analysis, learning guidance, and career development support within a single system. Such a platform should be able to analyze educational qualifications, technical skills, certifications, internships, projects, and resume content to generate meaningful insights and recommendations tailored to individual users. The system should also help students understand market demands and prepare effectively for placement opportunities.

The proposed Edu2Job – AI Career Intelligence System fulfills these requirements by combining Machine Learning, Natural Language Processing, Generative AI, FastAPI, React, and MongoDB technologies into a unified platform. The system predicts suitable career roles, analyzes resumes, identifies missing skills, generates personalized career roadmaps, recommends courses, and provides administrative analytics. By integrating multiple intelligent modules into a single platform.

Chapter 3

DEVELOPMENT METHODOLOGY

Starting with raw inputs, the Edu2Job setup grows step by step using smart algorithms alongside web tools to shape a thinking-style career helper. Instead of guessing, it pulls in personal details, shapes them into patterns, then feeds those into learning machines built to understand job paths. From there, resume content gets broken down piece by piece, letting the system spot missing abilities quietly. After spotting what's lacking, suggestions appear - tailored, quiet, fitting just one person at a time - all held together under a single working design.

From start to finish, building the system moves through several steps - first comes understanding needs, then gathering data, cleaning it up, shaping useful traits, crafting a learning model, connecting server tools, designing user screens, checking performance, finally mapping how it launches. Inside Edu2Job, parts like job forecasting, resume review, skill checking, smart coaching, and admin tools fit together like puzzle pieces

The Edu2Job platform is designed using a modular architecture where different components such as Job Prediction, Resume Analysis, Skill Gap Analysis, AI Career Guidance, and Admin Dashboard work together to provide a complete career intelligence solution.

The methodology used in this project not only focuses on technical implementation but also emphasizes scalability, usability, and real-world applicability. The integration of Artificial Intelligence and Machine Learning helps the platform provide personalized and intelligent career guidance according to the user's educational and technical profile.

3.1 Agile Software Development Methodology

Agile Software Development Methodology is one of the most widely used software development approaches in modern software engineering. Agile focuses on flexibility, continuous improvement, teamwork, and iterative development. Unlike traditional development models where the complete system is developed in a fixed sequence,

Agile divides the project into smaller development phases called iterations or sprints. Each sprint focuses on developing, testing, and improving specific functionalities of the application.

Because the Edu2Job – AI Career Intelligence System includes several linked parts - like predicting jobs, checking resumes, spotting skill gaps, offering career advice through AI, handling logins, and running an admin panel - it made sense to go with Agile. Working piece by piece let adjustments happen smoothly, even mid-process. Changes to models, tweaks in how things look, upgrades to features - all fit naturally into sprints instead of derailing progress. The method held up well when updates piled on, keeping everything moving without freezing the whole setup.

Step one broke the project into chunks, then work moved piece by piece. First up: screens people see, along with the hidden connections that power them. Next came smart guess engines, tools to scan resumes, followed by artificial brain bits slotted in slowly. Each chunk got checked on its own, tweaked until solid, only after did they join the main build. Building it like layers made tangled problems easier, kept everything running smoother in the long run.

Testing never stopped under Agile, woven into every part of the work. When a sprint ended, tools like prediction engines or login systems got checked right away. Instead of waiting, teams hunted down bugs in features such as resume parsing or data dashboards. Catching glitches early kept small hiccups from turning into roadblocks further on. Each round of checks tightened performance before moving ahead. One big plus of using Agile here? Better teamwork and clearer talk while building things. Frontend work, backend systems, machine learning bits, along with artificial intelligence parts - each moved together because tech areas stayed in step.

Another major advantage of Agile methodology in this project was improved collaboration and communication during development. Different components such as frontend design, backend APIs, Machine Learning integration, and AI modules required coordination between multiple technologies. Agile development allowed continuous monitoring of project progress and easier integration of new features whenever required. The methodology also helped improve the user experience of the platform.

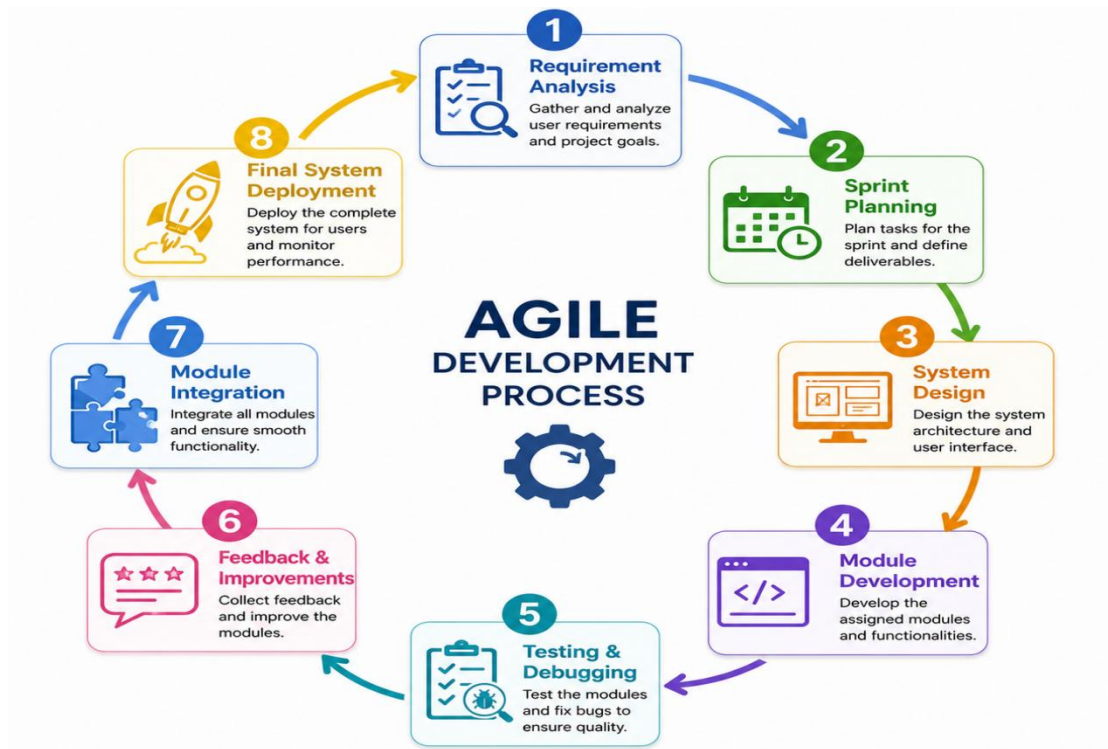


Fig. 3.1: Agile Development Process

Agile methodology additionally allowed faster and more manageable development because working functionalities could be completed independently. Instead of waiting for the entire project to finish, modules such as authentication, prediction engine, resume analyzer, and analytics dashboard were developed and tested separately. This made the development process more organized, flexible, and efficient.

Overall, Agile Software Development Methodology played an important role in the successful development of the Edu2Job platform. The methodology provided flexibility, continuous improvement, efficient testing, and modular development, which helped create a scalable and intelligent AI-powered career guidance system. As a result, the project development process became more manageable, flexible, and efficient.

3.2 Requirement Analysis Phase

Requirement analysis is the initial and one of the most important phases in the software development lifecycle. In this phase, the requirements, objectives, functionalities, and expectations of the system are identified, analyzed, and documented carefully. Requirement analysis helps developers clearly understand what

problems the software should solve, what features need to be implemented, and how the overall system should function. A proper requirement analysis phase reduces development confusion, improves project planning, and helps create a more efficient and reliable application.

Starting with Edu2Job – AI Career Intelligence System, the first stage looked closely at what students often find tough when picking careers. One big hurdle? Figuring out which jobs fit their studies, tech abilities, certificates, hands-on projects, or internship experience. Preparing for placements brings its own set of difficulties. Building a strong resume doesn't come easily to many. Skill growth paths can feel unclear without clear direction.

From watching how placements happen today, clues about what the project must do began to emerge. Recruitment patterns gave hints, while a look at present-day career tools added more depth. Student hopes, along with school demands, shaped part of the picture too. Instead of just listing wants, real needs took form - like guessing future jobs or checking resumes. Spotting missing skills led to ideas for features that respond personally. Preparation support for hiring rounds became another quiet pillar. Each function grew from actual behavior, not guesses.

Among what the system needed: signing users up, checking identities, guessing job fits. Uploading resumes came next, then breaking them down piece by piece. Career advice powered by artificial intelligence shaped one part. Spotting missing abilities mattered too. Suggestions for courses followed naturally. Admins got tools to run things behind the scenes. Charts showed patterns clearly. Connecting outside systems completed it. Each feature fit together so students and staff get clear direction without gaps.

The system also required several non-functional requirements such as security, scalability, reliability, responsiveness, usability, and performance. Since the platform stores user profiles, resumes, and prediction data, security was considered an important requirement during development. JWT-based authentication and secure API communication mechanisms were selected to protect user information and provide controlled system access.

Scalability was another important requirement because the system architecture needed to support future expansion, additional AI features, and large numbers of users.

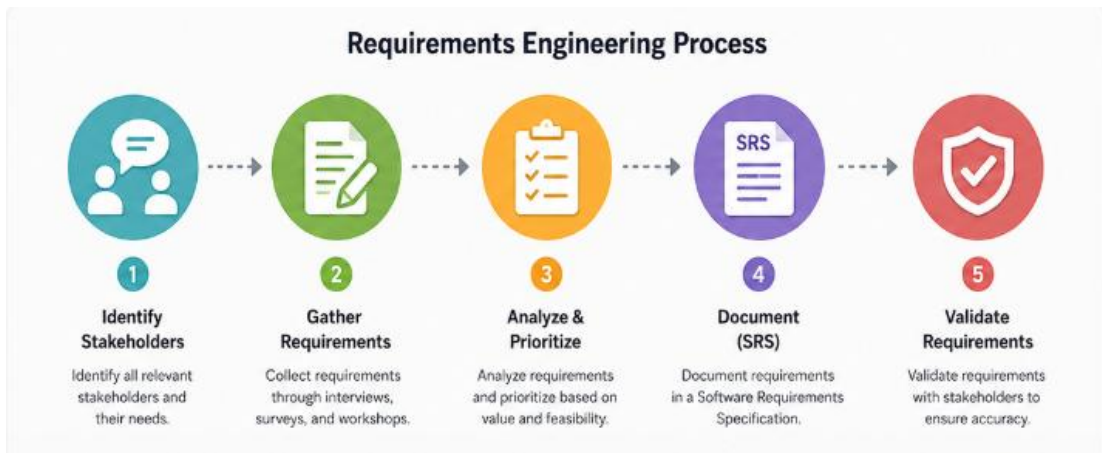


Fig. 3.2: Requirement Analysis Phases

Starting off, looking at what was needed made it clearer how big the whole project would be. That step shaped how data would be stored, along with how parts of the app talk to each other behind the scenes. Communication between user interface and server took form here, guided by early decisions. The design of interfaces that connect systems emerged naturally from these talks. Even the way machine learning tasks run got outlined during this time. As things moved forward, tools for building the visible part of the system locked into place. What runs on servers settled next, followed by choices about storing information. Smart algorithms found their spot once priorities became clear. Picking methods for training models happened last, fitting around earlier picks.

Overall, the requirement analysis phase played a major role in the successful development of the Edu2Job platform. It provided a clear understanding of user expectations, technical requirements, project objectives, and system workflows, which helped reduce development risks and improve the quality of the final application.

3.3 System Design Phase

What happens after gathering needs? Design shapes those inputs into clear plans for building software. Built right, it maps out how everything connects - structure, data flow, features, touchpoints - with precision. Instead of jumping into code, time spent here prevents problems later. For Edu2Job – AI Career Intelligence System, thought went deep into making a framework that grows smoothly with demand. Machine learning functions had to fit naturally within the setup. Resume scanning by artificial intelligence needed room to adapt without breaking rhythm elsewhere. Finding

missing skills required logic paths designed early. Personal advice engines were threaded through interaction models from the start. Dashboards showing insights relied on backend flows shaped during this stage. Each piece got attention so nothing felt tacked on once coding started. Blueprints drawn now guide every line written later. Built on three levels, Edu2Job splits its structure into Frontend, Backend, and a combined Database & AI section. Logic rules, service links, login checks, plus ML features run through the middle piece.

During system design, plans took shape around adding smart tools like AI and machine learning. Instead of simple rules, decision paths grew using a Random Forest Classifier tucked inside the back end to guess job matches. On top of that, reading resumes became possible by weaving in Natural Language Processing along with Gemini AI for tailored advice on careers. To sketch how everything fits, diagrams popped up - Use Case ones showed who does what, Sequence ones traced steps between parts. Each drawing peeled back layers on how users engage, how pieces talk, how actions flow across the platform. Safety stayed baked in throughout these blueprints.

- **Use Case Diagram**

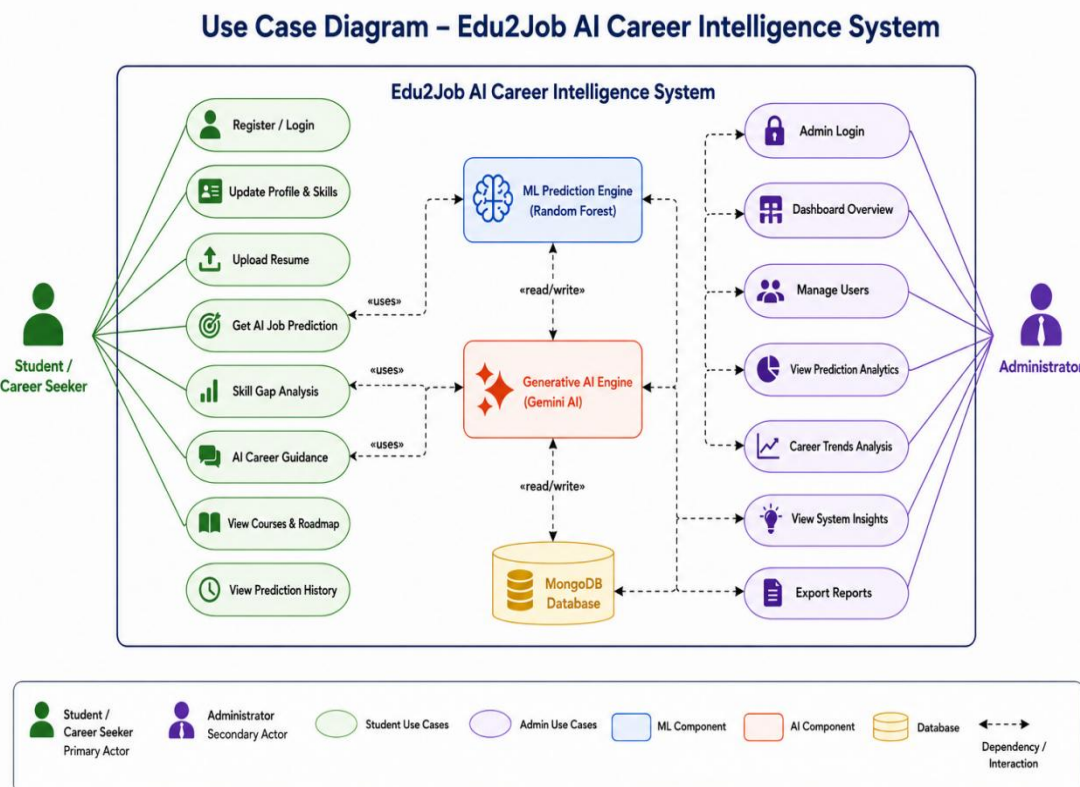


Fig. 3.3: Use Case Diagram

The Use Case Diagram represents the functional overview of the Edu2Job AI Career Intelligence System and illustrates the interaction between users and system components. The diagram mainly includes two actors: Student/Career Seeker and Administrator. It also shows the interaction of the platform with supporting components such as the Machine Learning Prediction Engine, Generative AI Engine, and MongoDB Database.

The diagram provides a high-level view of how users access various services and functionalities offered by the platform. The primary actors involved in the system are Student/User and Administrator. The Student is the main user of the platform and interacts with multiple modules designed to support career development and employability enhancement. Initially, users can register and log in to the platform through a secure authentication mechanism. After successful authentication, users gain access to their personalized dashboard, where they can manage their career development activities.

One of the primary functionalities available to users is the AI Job Prediction Module. Students can enter their academic details, technical skills, certifications, internship information, projects, and career interests. The system processes this information using the Machine Learning prediction engine and generates suitable career role recommendations. The predicted roles help users understand which career paths best align with their educational background and skill profile. Users can also access the Resume Analyzer Module, where they upload resumes for intelligent evaluation.

Another important functionality is Skill Gap Analysis, which compares a user's existing skills with industry requirements for target career roles. The system identifies missing competencies and recommends technologies, certifications, tools, and learning resources required to improve employability. This helps users focus on acquiring relevant skills demanded by the industry.

The platform additionally provides a Course Recommendation Module, where users receive learning resources and certification suggestions based on prediction results and identified skill gaps. These recommendations support continuous learning and professional growth.

● Sequence Diagram

The The Sequence Diagram illustrates the step-by-step interaction between the Student, Frontend Interface, Backend API, Machine Learning Model, and MongoDB Database during the AI Job Prediction process. It represents how the system processes user input and generates intelligent career predictions.

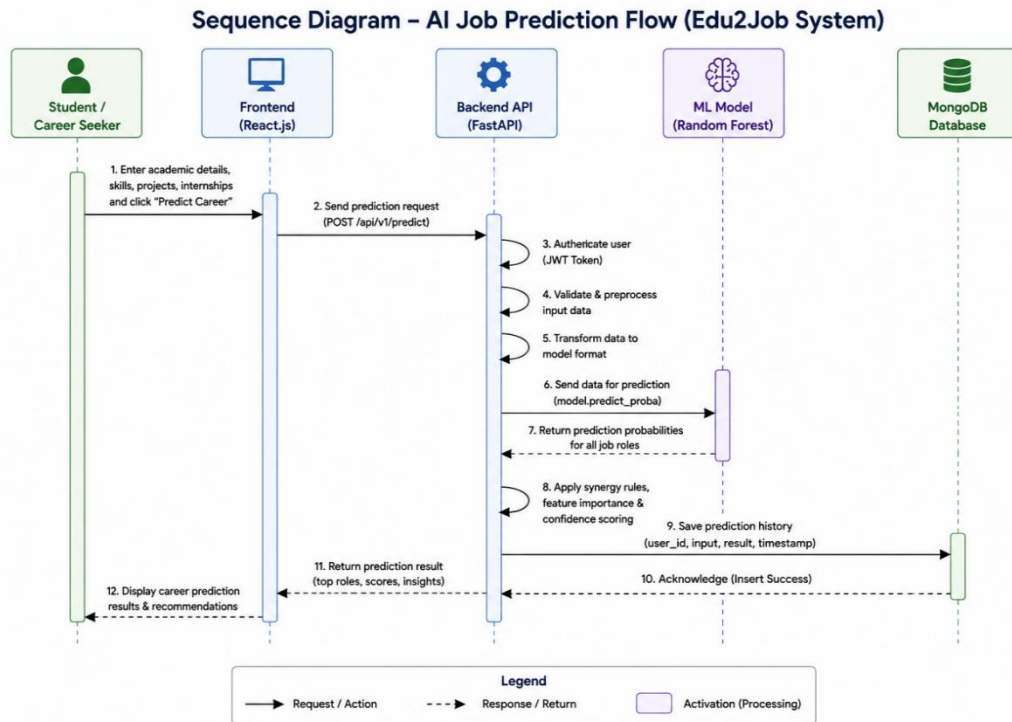


Fig. 3.4: Sequence Diagram

The process begins when the student enters academic details, technical skills, projects, internships, and other profile information through the React.js frontend interface. After filling in the details, the student clicks on the “Predict Career” option. The frontend then sends the prediction request to the FastAPI backend through REST APIs. The backend first authenticates the user using JWT-based authentication and validates the received data. After validation, preprocessing operations such as data transformation, feature extraction, and formatting are performed to convert the user input into a Machine Learning-compatible structure.

The processed data is then forwarded to the Gradient boosting Machine Learning model. The model analyzes the input features and generates

prediction probabilities for multiple job roles. Once the prediction analysis is completed, the backend stores the prediction history and related information inside the MongoDB database. After successful database insertion, the backend sends the final prediction results back to the frontend interface, displayed to the user as ranked career recommendations along with skills analysis and career insights.

For the Resume Analyzer workflow, users upload their resumes through the frontend interface. The uploaded file is transferred to the backend, where Natural Language Processing techniques are applied to extract educational qualifications, technical skills, certifications, projects, and experience. The extracted information is stored temporarily and passed to the prediction and recommendation modules.

Overall, the Sequence Diagram demonstrates the collaboration between the React frontend, FastAPI backend, MongoDB database, Machine Learning prediction engine, NLP processing module, and Generative AI services. It highlights the complete flow of information and illustrates how multiple intelligent technologies work together to provide personalized career guidance and employability enhancement.

3.4 Dataset Collection and Preparation

Well the model predicts things ties closely to how good the data looks and how it is set up. For Edu2Job, the team pulled together details often linked to hiring and job placements - focusing on school results, tech abilities, hands-on projects, certificates, past intern roles, field of study, and parts taken from resumes. What ended up inside the dataset reflects real-world traits companies tend to check. From education records to hands-on training, the data behind Edu2Job – AI Career Intelligence System covers many sides of a learner’s journey.

One piece shows what degree someone pursued, along with their area of focus during studies. Another reveals how well they performed overall, using scores like CGPA as a measure. What they learned in key topics tied to their major also plays into the picture. Skills in coding and tech tools appear next, giving insight beyond classrooms. Separate entries track which programming languages the person can handle. Proof of extra learning appears through certificates earned outside required

courses. Real-world exposure comes through past internships listed in each profile. Projects finished during school time round out the full view of ability

From resume details pulled using language processing tools, the data expands past what students type themselves. Because project summaries, skill sets, licenses, and hands-on work appear here, clear signals emerge about ability and preparedness. Output targets show up too - these mark the jobs learners aim for when teaching the algorithm. When connections form between those traits and actual role choices, pattern detection sharpens inside the Gradient Boosting method. Accuracy climbs since personal education paths, talents, and background shape every suggested opportunity uniquely.

Sorted details landed in clean rows and columns, helping steps before machine learning get smoother. Because actual job history records often stay hidden - locked by access limits or personal secrecy - made-up sets joined hand-picked examples to mirror how students really move through careers. Then patterns began showing up.

The collected data was organized into structured tabular format for easier preprocessing and Machine Learning training. Since real-world career datasets are difficult to obtain due to privacy and availability constraints, synthetic and manually curated datasets were also used to simulate realistic student profiles and placement scenarios.

3.5 Data preprocessing

Data preprocessing is an important step in Machine Learning because raw data often contains inconsistencies, missing values, unstructured text, and categorical information that cannot be directly processed by ML algorithms.

In the Edu2Job project, preprocessing techniques were applied to convert raw educational and technical data into a structured format suitable for Machine Learning training and prediction. Categorical data such as degree names, specialization, certifications, and technical skills were converted into numerical representations using encoding techniques. Text-based information such as projects, skills, and resume keywords were processed using Natural Language Processing techniques and vectorization methods. Missing or incomplete values were handled carefully to maintain dataset consistency and avoid prediction errors. Academic scores and

technical skill weights were normalized to improve model performance and balance feature importance.

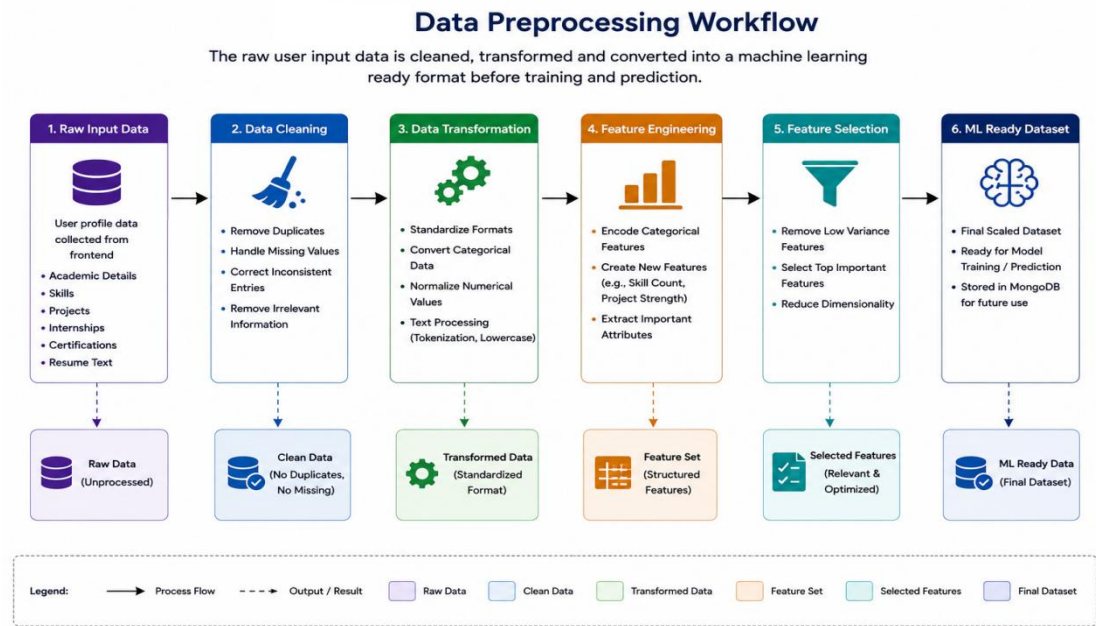


Fig. 3.5:Data Preprocessing Workflow

Feature extraction techniques were also applied to identify important keywords and technical patterns from resumes and project descriptions. These extracted features helped improve prediction accuracy and career role matching. The preprocessing stage ensured that the dataset became clean, structured, and optimized for Machine Learning model training.

TECHNOLOGY STACK

Technology stack refers to the collection of programming languages, frameworks, libraries, databases, APIs, and software tools used for developing a software application. Selecting an appropriate technology stack is one of the most important aspects of software development because it directly affects system performance, scalability, security, maintainability, and user experience. A well-designed technology stack helps developers build reliable and efficient applications capable of handling real-world operations smoothly.

The Edu2Job – AI Career Intelligence System was developed using modern full-stack web technologies, Artificial Intelligence frameworks, Machine Learning libraries, and cloud-based AI services to create an intelligent and scalable career guidance platform. Since the project includes functionalities such as AI-based job prediction, resume analysis, skill-gap detection, analytics dashboards, authentication, and recommendation systems, it required a robust and flexible technology stack capable of handling both frontend and backend operations efficiently.

The frontend of the application was developed using React.js along with HTML, CSS, and JavaScript. React.js was selected because it supports component-based architecture, reusable UI components, fast rendering, and responsive web application development. The frontend is responsible for handling user interaction, displaying dashboards, prediction results, analytics charts, and AI-generated recommendations.

The backend of the project was implemented using FastAPI and Python programming language. FastAPI is a high-performance backend framework used for creating RESTful APIs and integrating Machine Learning models efficiently. The backend handles functionalities such as authentication, API communication, Machine Learning prediction processing, resume analysis, AI integration, and database management. MongoDB was selected as the database management system because the project stores semi-structured and dynamic data such as resumes, prediction history, analytics records, and AI-generated recommendations. MongoDB provides scalability,

flexibility, and efficient handling of JSON-like data structures, making it suitable for AI applications.

Fig. 4.1: Technology Stack Workflow of Edu2Job

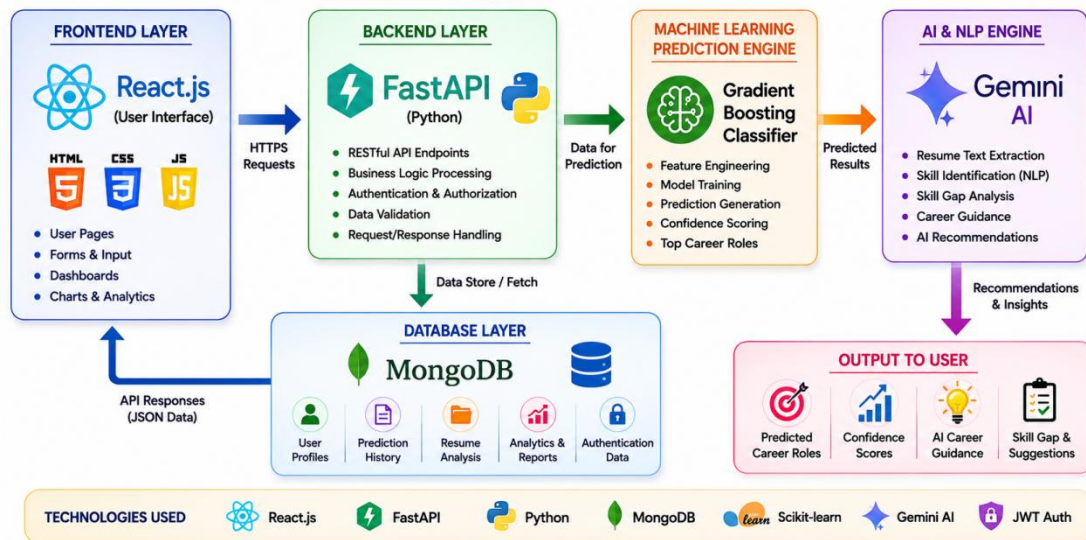


Fig. 4.1: Technology Stack Workflow

JWT-based authentication mechanisms were implemented to provide secure login functionality and role-based authorization for students and administrators. RESTful APIs were additionally used to establish communication between frontend interfaces, backend services, Machine Learning modules, and AI systems. architecture.

4.1 Frontend Technologies

Frontend technologies are responsible for designing the user interface and improving interaction between users and the application. The frontend of the Edu2Job – AI Career Intelligence System was developed using React.js along with HTML, CSS, and JavaScript.

The frontend plays a major role in providing an interactive and responsive environment where students can enter profile details, upload resumes, view career predictions, analyze skill gaps, and access AI-generated recommendations. Since the platform contains multiple dashboards, analytics sections, and AI-based functionalities, it required a modern frontend framework capable of handling dynamic content efficiently.

React.js was selected as the primary frontend framework because it supports component-based architecture, reusable UI components, fast rendering, and efficient

state management. React.js allows developers to divide the application into smaller reusable components, which improves code organization, scalability, maintainability, and frontend performance. The component-based structure also simplified the development of multiple modules such as Login, Dashboard, Resume Analyzer, Job Prediction, Skill Gap Analysis, and Admin Analytics.

HTML was used for structuring web pages and creating user interface elements such as forms, buttons, dashboards, tables, charts, and navigation menus. CSS was used for styling the application, creating responsive layouts, improving visual appearance, and enhancing overall user experience.

JavaScript was used for implementing frontend logic, event handling, API communication, form validation, and dynamic content rendering. React.js internally uses JavaScript for creating interactive user interfaces and handling component behavior. JavaScript additionally helped improve frontend responsiveness and enabled real-time interaction between frontend components and backend APIs.

The frontend technologies helped create responsive dashboards for students and administrators. Different functionalities such as job role prediction, resume upload, AI career guidance, analytics visualization, skill-gap reporting, and prediction history management were implemented using React.js components and frontend services. The frontend also supports real-time API communication with the FastAPI backend for displaying Machine Learning predictions and AI-generated recommendations dynamically.

Overall, the frontend technologies used in the Edu2Job project helped create a modern, scalable, responsive, and user-friendly AI-powered career intelligence platform capable of providing smooth interaction and efficient visualization of prediction and analytics results.

4.2 React.js Framework

React.js is the primary frontend framework used for developing the Edu2Job – AI Career Intelligence System. React.js is one of the most popular JavaScript libraries used for building modern single-page applications because it provides fast rendering, reusable components, virtual DOM architecture, and efficient frontend performance. This modular development approach improved code organization, frontend maintainability, scalability, and development efficiency. Since every module

was implemented independently, debugging and future enhancements became easier during development.

React routing was implemented for managing navigation between different pages and dashboards. This allowed smooth page transitions without reloading the complete application. Dynamic routing additionally helped provide role-based access for students and administrators according to authentication permissions. React Hooks and state management techniques were used for handling user data, API responses, form validation, prediction results, and dynamic UI updates. These features improved frontend responsiveness and enabled real-time interaction between frontend components and backend APIs.

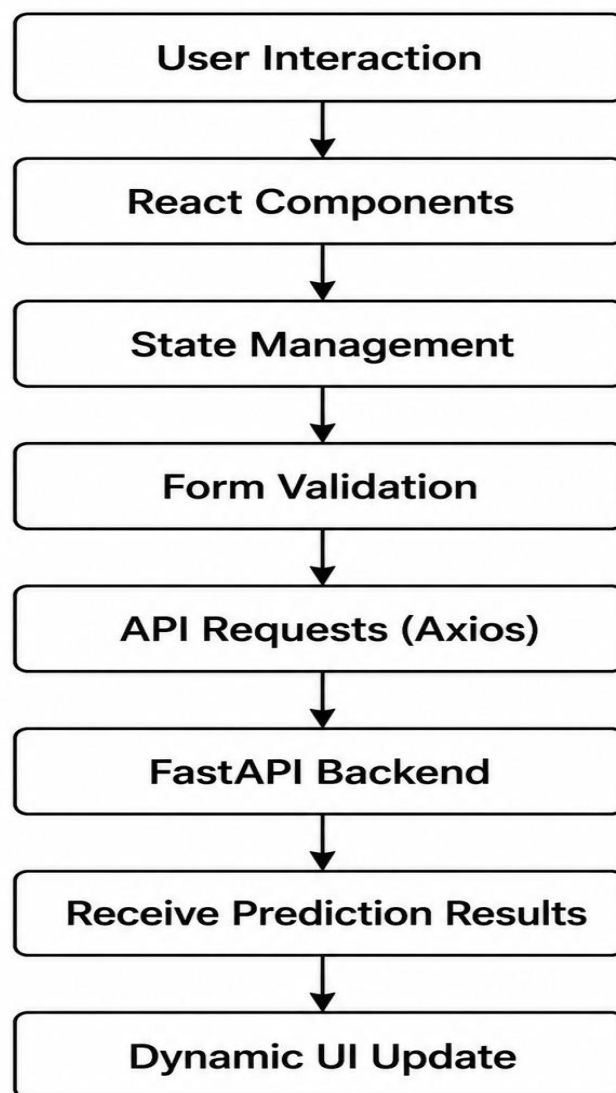


Fig. 4.2: React.js Workflow

Charts, graphs, score indicators, analytics cards, and visual prediction panels were integrated using frontend libraries to improve the presentation of career insights and analytics information. The frontend design additionally focused on responsive web development so that the application could function properly across desktops, tablets, and mobile devices. CSS styling, grid systems, flex layouts, and responsive design techniques were used to improve visual consistency and accessibility..

The above figure illustrates the workflow of the React.js frontend framework used in the Edu2Job platform. React components handle user interaction, API communication, state management, and dynamic rendering of prediction and analytics results. Overall, React.js played a major role in building a modern, responsive, scalable, and interactive frontend architecture for the Edu2Job platform. The framework helped improve performance, user interaction, maintainability, and real-time communication with Machine Learning and AI backend services.

4.3 Backend Technologies

Backend technologies are responsible for processing business logic, handling authentication, managing database communication, integrating Machine Learning models, and controlling API communication between different modules of the application. The backend acts as the core processing layer of the Edu2Job – AI Career Intelligence System because it manages prediction workflows, resume analysis operations, AI integration, analytics processing, and secure communication between frontend and database systems.

The backend of the Edu2Job platform was developed using FastAPI framework and Python programming language. These technologies were selected because they provide high performance, scalability, flexibility, and smooth integration with Artificial Intelligence and Machine Learning libraries.

FastAPI is a modern Python-based backend framework used for developing high-performance RESTful APIs. FastAPI was selected because it supports asynchronous request handling, automatic API documentation, fast processing speed, and easy integration with Machine Learning models. The framework additionally simplifies API development and improves backend maintainability through modular architecture and structured routing systems.

Python programming language was selected because of its simplicity, readability, extensive AI ecosystem, and strong support for Machine Learning, Natural Language Processing, and data processing libraries. These APIs receive requests from frontend dashboards, process the data, interact with Machine Learning and AI modules, and return prediction results and analytics information back to the frontend interface.

The backend architecture was designed using modular development principles so that additional functionalities, Machine Learning models, and AI services can be integrated easily in future versions of the platform. This makes the Edu2Job system scalable and adaptable to changing technologies and industry requirements.

Overall, the backend technologies used in the Edu2Job project helped create a secure, scalable, intelligent, and high-performance AI-powered career guidance platform capable of handling real-time Machine Learning predictions, AI integration, analytics processing, and efficient frontend-backend communication.

4.4 Database Technology

MongoDB is the database management system used in the Edu2Job – AI Career Intelligence System. Modern AI-based applications generate large amounts of dynamic and semi-structured data such as resumes, prediction history, analytics records, recommendation outputs, and user activity logs. Therefore, the project required a flexible and scalable database system capable of handling structured as well as unstructured information efficiently.

MongoDB was selected because it provides high scalability, fast query performance, flexible document-based storage, and efficient handling of JSON-like data structures. Unlike traditional relational databases, MongoDB stores data in collections and documents, which makes it suitable for applications involving Artificial Intelligence, Machine Learning, and dynamic user-generated content.

The Edu2Job platform stores various types of information including user profiles, educational details, technical skills, certifications, project information, resume analysis results, prediction history, analytics data, and administrator records. MongoDB helped organize this information efficiently without requiring rigid relational table structures.

The Users collection stores information related to student accounts, educational details, technical skills, certifications, and authentication credentials. The Predictions collection stores career prediction history, confidence scores, and recommendation data generated by the Machine Learning model. Resume analysis results, extracted skills, missing skills, and AI-generated recommendations are stored inside the Resume Analysis collection.

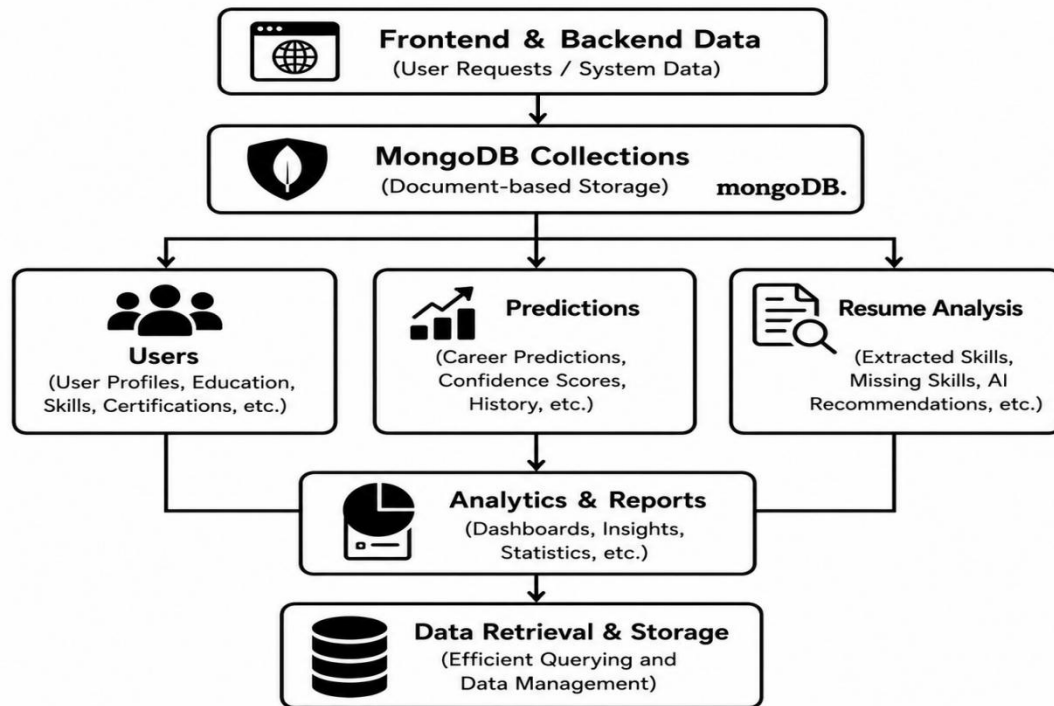


Fig. 4.3: MongoDB Workflow

Database indexing and query optimization techniques were implemented to improve retrieval speed and reduce response time during analytics processing and prediction history access. Proper validation mechanisms were also implemented to maintain data consistency and prevent invalid records from being stored in the database.

Another major advantage of MongoDB is scalability. As the number of users and prediction records increases, the database can easily support additional collections, analytics modules, and AI-generated datasets without major architectural changes. This makes the Edu2Job platform suitable for future expansion and large-scale deployment. Overall, MongoDB provided a scalable, flexible, and efficient database solution for the Edu2Job platform. The database technology helped manage prediction

data, AI-generated recommendations, analytics information, and user records effectively while supporting real-time interaction between frontend, backend, and Machine Learning systems.

4.5 Authentication and Security Technologies

Authentication and security are among the most important aspects of the Edu2Job – AI Career Intelligence System because the platform stores sensitive user information such as resumes, educational details, prediction history, analytics data, and career recommendations.

The Edu2Job platform uses JWT (JSON Web Token) authentication for secure login functionality and API authorization. JWT is a token-based authentication mechanism that allows secure communication between frontend and backend systems without repeatedly storing session information on the server.

During the login process, users enter their credentials through the frontend interface. The backend validates the credentials using FastAPI authentication APIs and generates a JWT authentication token for successfully authenticated users. This token is then stored on the client side and attached to future API requests for verification. Whenever protected API routes are accessed, the backend validates the JWT token before processing the request. If the token is valid, the requested operation is allowed; otherwise, access is denied.

Overall, JWT authentication, role-based authorization, secure password management, and protected API communication helped create a secure and reliable AI-powered career intelligence platform capable of protecting sensitive user data and backend services.

4.6 API Integration Technologies

API integration technologies play an important role in the Edu2Job – AI Career Intelligence System because the platform depends on communication between frontend interfaces, backend services, Machine Learning modules, AI systems, and database services. APIs allow different components of the application to exchange information efficiently and perform real-time operations.

The Edu2Job platform uses RESTful APIs developed using FastAPI for communication between the React.js frontend and backend services. These APIs

process requests related to user authentication, career prediction, resume analysis, skill-gap analysis, analytics generation, recommendation systems, and admin dashboard operations.

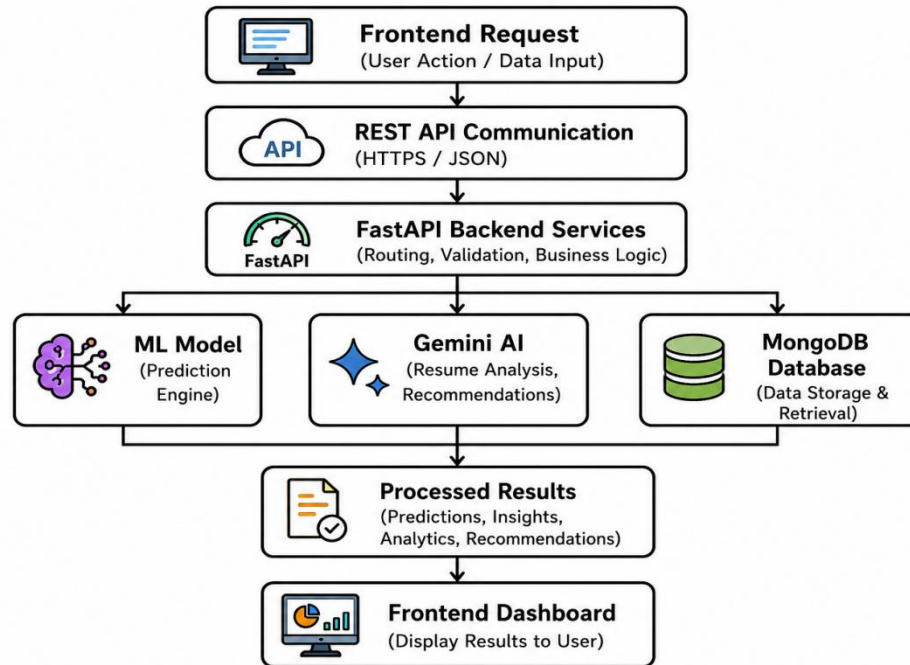


Fig. 4.4: API Workflow

When users enter profile details or upload resumes through the frontend interface, the information is sent to backend APIs for processing. The backend APIs then interact with Machine Learning models, Natural Language Processing modules, Gemini AI services, and MongoDB database before returning the final prediction results and recommendations to the frontend dashboard.

RESTful API architecture simplified frontend-backend communication and improved system scalability. Since APIs follow standardized request-response mechanisms, additional services and functionalities can easily be integrated into the platform in future versions. The API integration architecture additionally supports future expansion of the platform.

Overall, API integration technologies helped establish smooth communication between frontend interfaces, backend services, Machine Learning modules, AI systems, and database layers.

4.7 Machine Learning and Artificial Intelligence Technologies

The Edu2Job system uses a Machine Learning-based prediction engine to identify suitable career roles according to the user's profile. Different Machine Learning algorithms were studied during development, including Decision Trees, Support Vector Machines, Naive Bayes, Random Forest, and Gradient Boosting techniques.

After comparing accuracy, scalability, prediction performance, and model reliability, the Gradient Boosting Classifier was selected as the primary prediction model because of its high accuracy and strong ability to handle complex structured datasets efficiently. The model performed well in identifying relationships between technical skills, educational qualifications, certifications, internships, and career roles.

Gradient Boosting is a supervised Machine Learning algorithm that works by combining multiple weak learners, usually decision trees, to create a strong prediction model. Unlike Random Forest, where trees work independently, Gradient Boosting builds trees sequentially. Each new tree attempts to correct the errors made by the previous trees, which improves overall prediction accuracy and model performance.

The model was trained using processed educational and technical datasets prepared during the data preprocessing stage. During training, the system learned patterns and relationships between features such as programming skills, project technologies, certifications, academic scores, internship experience, and target career roles.

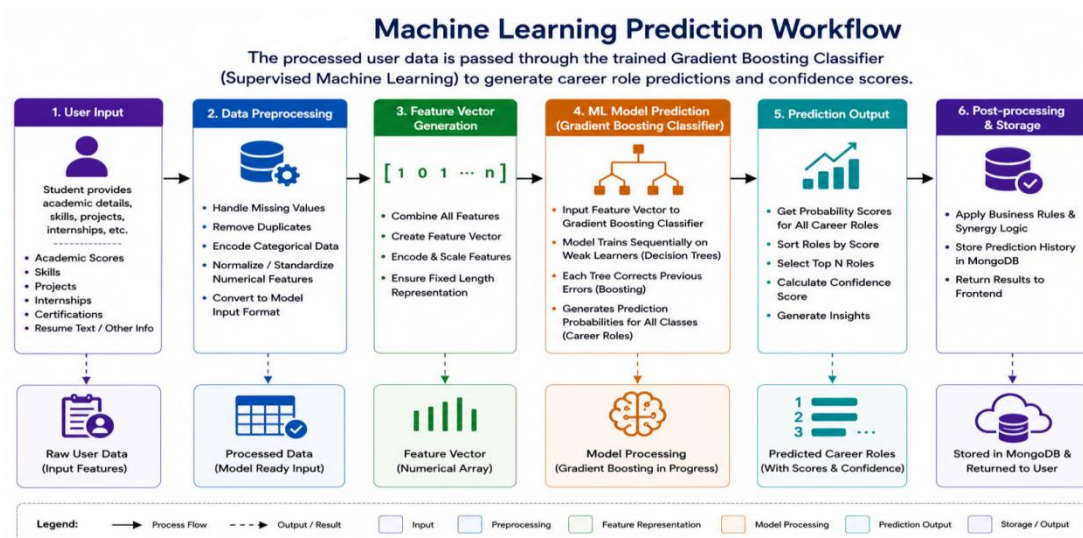


Fig. 4.5: Machine learning Prediction Workflow

The trained Gradient Boosting Classifier model was then integrated into the backend system using FastAPI APIs for real-time prediction generation. Whenever a user enters profile details into the platform, the backend preprocesses the data and sends it to the trained Machine Learning model, which then predicts the most suitable career roles along with confidence scores and recommendation insights.

4.7.1 Machine Learning-Based Career Prediction

The primary objective of the Machine Learning engine is to predict the most suitable career role for a student based on academic performance, educational background, technical skills, project experience, internship experience, and subject knowledge. The system uses a Gradient Boosting Classifier, an ensemble learning algorithm that combines multiple decision trees to generate highly accurate classification results. The model is trained to classify student profiles into multiple career categories and provide ranked career recommendations according to profile suitability.

The model was trained using a domain-aligned dataset containing more than 8,000 student profiles distributed across major domains such as Artificial Intelligence, Software Development, Civil Engineering, Finance, and Management. Each record contains categorical attributes such as degree and specialization, numerical attributes such as academic scores and experience, and textual information including skills, projects, internships, and core subjects. This diverse dataset enables the model to learn relationships between educational qualifications and career outcomes effectively.

The core objective of the Edu2Job ML engine is Career Path Prediction. The engine evaluates a student's academic profile—comprising degrees, grades, standardized marks, specialized skills, projects, and internships—and matches them to the most aligned career roles.

The architecture comprises three main components:

- 1) NLP Resume Parser:** Extracts profile features from raw resume text using Spacy.
- 2) Machine Learning Classifier:** A Scikit-Learn pipeline using Gradient Boosting to classify profiles into 24 target career roles.

- 3) **GenAI Enhancer:** Integrated Google Gemini 1.5 Flash services for career roadmapping, gap simulation, and project recommendation.

4.7.2 Feature Engineering and Data Processing

Before training, the collected data undergoes extensive preprocessing and feature engineering. A custom feature transformation pipeline extracts meaningful information from raw student data. Academic subject grades are converted into numerical values and aggregated to calculate average subject performance. Internship durations are extracted and transformed using logarithmic scaling to normalize experience values. Project counts are derived from project descriptions, while skills, subjects, and project information are combined into a consolidated text representation for further analysis.

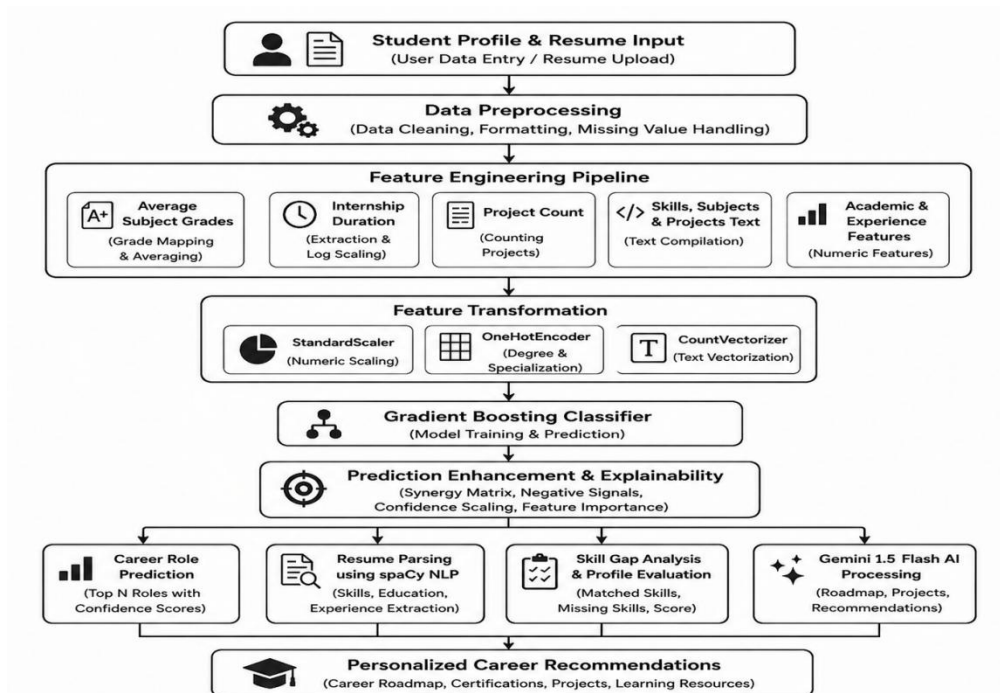


Fig. 4.6:Machine learning and AI Workflow

Component Details:

1) Custom Feature Transformer:

- **Academic Subject Grades:** Maps grade characters to numeric values using a predefined map (A+/O/S = 6, A = 5, down to C = 1, D = 0.5) and calculates the mean (avg_subject_grade).

- **Internship Duration:** Extracts durations (in months) using regex, sums them, and applies a logarithmic transformation:

$$\text{log_internship_duration} = \log(1 + \text{duration_months})$$
 - **Project Count:** Counts the number of semicolon-separated projects.
- 2) **Feature Scaler:** Applies StandardScaler to all numerical features.
 - 3) **Categorical Encoder:** Encodes degree and specialization using OneHotEncoder(handle_unknown='ignore').
 - 4) **Text Vectorizer:** Extracts text features using CountVectorizer

4.7.3 Model Configuration

The prediction engine utilizes the Scikit-Learn implementation of the Gradient Boosting Classifier. The model is configured with 150 estimators, a learning rate of 0.08, maximum tree depth of 5, and a subsampling rate of 0.8. During training, the dataset is divided into training and testing subsets using an 85:15 ratio. After successful training and evaluation, the complete Machine Learning pipeline is stored as a reusable model artifact and integrated with the backend prediction APIs for real-time career prediction.

4.7.4 Integration and Post-Prediction Rules

During backend inference (backend/api/predict.py), the raw predictions generated by the classifier are augmented with domain-specific rules to improve accuracy and explainability:

- 1) **Synergy Bonus Matrix:** To prioritize performance in critical academic courses, the backend uses a predefined subject-to-job map (SYNERGY_MATRIX):
 - If a student earns an A+, A, or A- in a subject matching a predicted role (e.g., "Machine Learning" grade A+ and predicted role "AI Engineer"), the probability score for that role receives a 1.3x scaling boost.

- 2) **Negative Signals** : Conversely, low grades (e.g., C or D) in foundational subjects (e.g., "Data Structures and Algorithms") penalize the probability score for high-end technical roles (e.g., "Software Engineer").
- 3) **3. Confidence Scaling**: Raw output probabilities from ensemble trees are often heavily skewed. The system normalizes these outputs using a custom scaling function to map raw probabilities into a human-friendly confidence range (e.g., 60% to 95%).

4.7.5 Testing and Validation

The Machine Learning and Artificial Intelligence modules of the Edu2Job system were thoroughly tested and validated to ensure prediction accuracy, reliability, and overall system performance. The Gradient Boosting Classifier was evaluated using training and testing datasets to verify its ability to predict suitable career roles based on academic records, technical skills, certifications, projects, and internships. The feature engineering pipeline, including data preprocessing, categorical encoding, numerical scaling, and text vectorization, was also tested to ensure that user data was correctly transformed into machine-readable features. Multiple student profiles were used during testing, and the model consistently generated meaningful career predictions with appropriate confidence scores.

The Resume Analyzer, Skill Gap Analysis, and Generative AI modules were additionally validated using various user scenarios and resume formats. The spaCy NLP engine successfully extracted skills, education details, projects, and experience from uploaded resumes, while the Skill Gap Analysis module accurately identified missing competencies and improvement areas. The Gemini AI module generated personalized career roadmaps, certification recommendations, project suggestions, and learning plans based on user profiles and target career goals. The successful testing results confirmed that the Edu2Job platform provides reliable career predictions, effective resume analysis, and personalized career guidance suitable for real-world deployment.

Chapter 5

IMPLEMENTATION DETAILS AND EXPERIMENTAL RESULTS

Implementation is one of the most important phases in software development because it converts the designed architecture, Machine Learning models, Artificial Intelligence modules, and system workflows into a fully functional software application. During the implementation phase, all modules of the Edu2Job – AI Career Intelligence System were developed, integrated, tested, and deployed successfully. The implementation process involved frontend development, backend API creation, Machine Learning model integration, resume analysis implementation, AI career guidance integration, database management, authentication setup, and analytics dashboard development.

The Edu2Job platform was implemented using modern technologies such as React.js, FastAPI, MongoDB, Gemini AI, Natural Language Processing, JWT Authentication, and the Gradient Boosting Classifier Machine Learning model. The implementation process focused on building a scalable, intelligent, responsive, and user-friendly platform capable of helping students make better career decisions based on their educational background, technical skills, certifications, projects, and professional interests.

The implementation phase additionally included integration between frontend interfaces, backend APIs, Machine Learning services, Artificial Intelligence modules, and database systems. RESTful APIs were developed to establish communication between different layers of the application. Proper API integration ensured smooth data exchange and real-time prediction generation throughout the platform.

The project was developed using Agile Software Development Methodology. Different modules such as user authentication, AI job prediction, resume analyzer, skill gap analysis, AI career guidance, analytics dashboards, and administrative monitoring systems were implemented incrementally and tested continuously. This development approach reduced implementation risks and improved software quality.

5.1 Home Page

Figure 5.1 represents the home page of the Edu2Job platform. The Home Page acts as the entry point of the Edu2Job – AI Career Intelligence System. This page introduces users to the platform and provides an overview of its major functionalities. The homepage was designed using React.js, HTML, CSS, and JavaScript to create a modern, responsive, and visually appealing interface. The page contains navigation options such as Home, About, Features, Contact, Login, and Register, allowing users to access different sections of the platform easily.

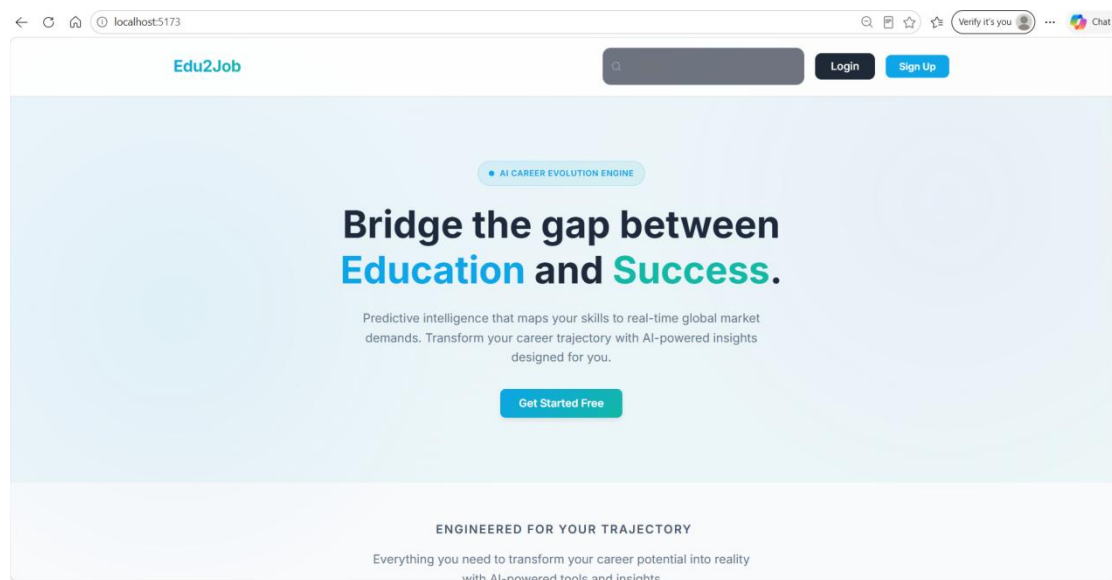


Fig. 5.1: Home Page

The homepage highlights the core features of the system including AI Job Prediction, Resume Analysis, Skill Gap Analysis, Career Roadmaps, and AI Career Guidance. Interactive sections and feature cards help users understand the benefits of the platform before accessing the prediction modules.

5.2 User Dashboard

The User Dashboard serves as the central control panel of the Edu2Job platform. After successful authentication, users are redirected to this page where they can access a complete overview of their career progress, skill readiness, and prediction results. The dashboard provides important information in a visually appealing format through cards, charts, and analytical components.

The dashboard displays several key indicators including predicted career role, match score, average salary, demand growth percentage, and top identified skills.

These values are generated using Machine Learning algorithms and analytics services integrated within the system.

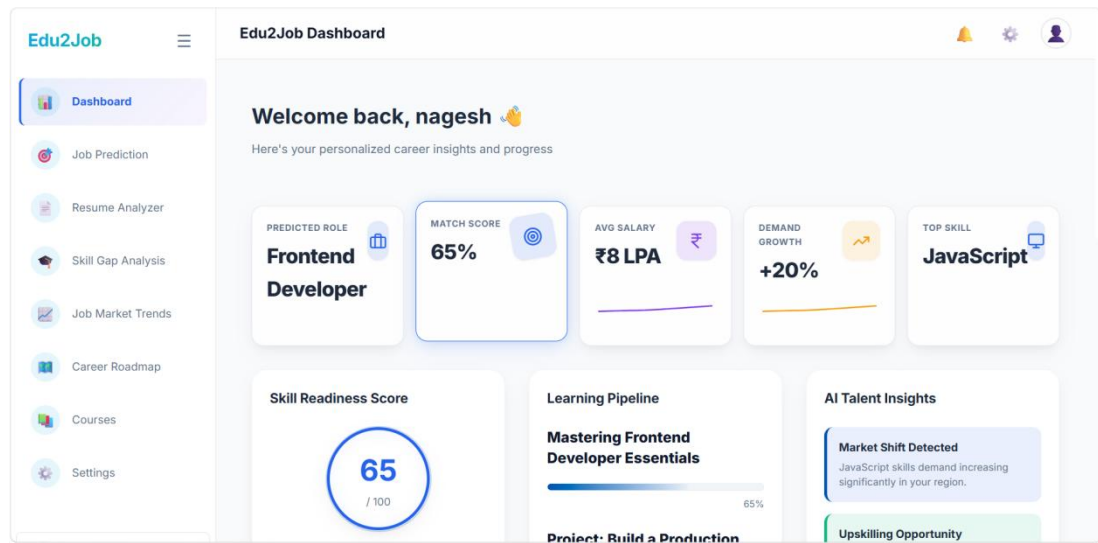


Fig. 5.2: User Dashboard

A Skill Readiness Score is also displayed, providing a numerical representation of the user's preparedness for their predicted career role. Additional sections such as Learning Pipeline and AI Talent Insights provide personalized recommendations and industry trends that can help users improve their employability.

5.3 AI Job Prediction Module – Academic Background, Skills & Experience, and Prediction Results

The AI Job Prediction Module is one of the core functionalities of the Edu2Job – AI Career Intelligence System. This module uses a trained Gradient Boosting Classifier Machine Learning model to analyze a student's educational background, technical skills, experience, projects, internships, and career interests in order to predict the most suitable job roles. The prediction process is divided into three major stages: Academic Background, Skills & Experience, and Prediction Results. This multi-stage approach helps collect comprehensive user information and improves the accuracy of career recommendations.

The Academic Background section represents the first stage of the prediction process. In this stage, users are required to provide detailed educational information including degree type, specialization, CGPA, 10th and 12th percentage, and performance in core academic subjects. The system allows students to enter subject-

wise grades for important engineering and computer science subjects such as Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, and other relevant courses.

Fig. 5.3: AI Job Prediction (Academic Background)

The Skills and Experience section represents the second stage of the prediction workflow. This stage focuses on collecting practical and technical information that reflects the student's real-world capabilities. Users can select multiple technical skills such as Python, Java, Machine Learning, Data Structures, Algorithms, Web Development, Database Management, Cloud Computing, and many other technologies. In addition to technical skills, the system also gathers information regarding career interests. Students can choose domains such as Software Development, Machine Learning, Data Science, Cybersecurity, Cloud Engineering, Full Stack Development, Web Development, Mobile Application Development, and other specialized courses.

The module also collects professional information including years of experience, completed projects, internships, certifications, freelance work, and other practical achievements. These details provide insights into the student's hands-on exposure and industry readiness preferences.

Project and internship information is particularly important because it demonstrates the application of theoretical knowledge in real-world scenarios. The combination of technical skills, career interests, experience, projects, and internships creates a comprehensive profile of the user.

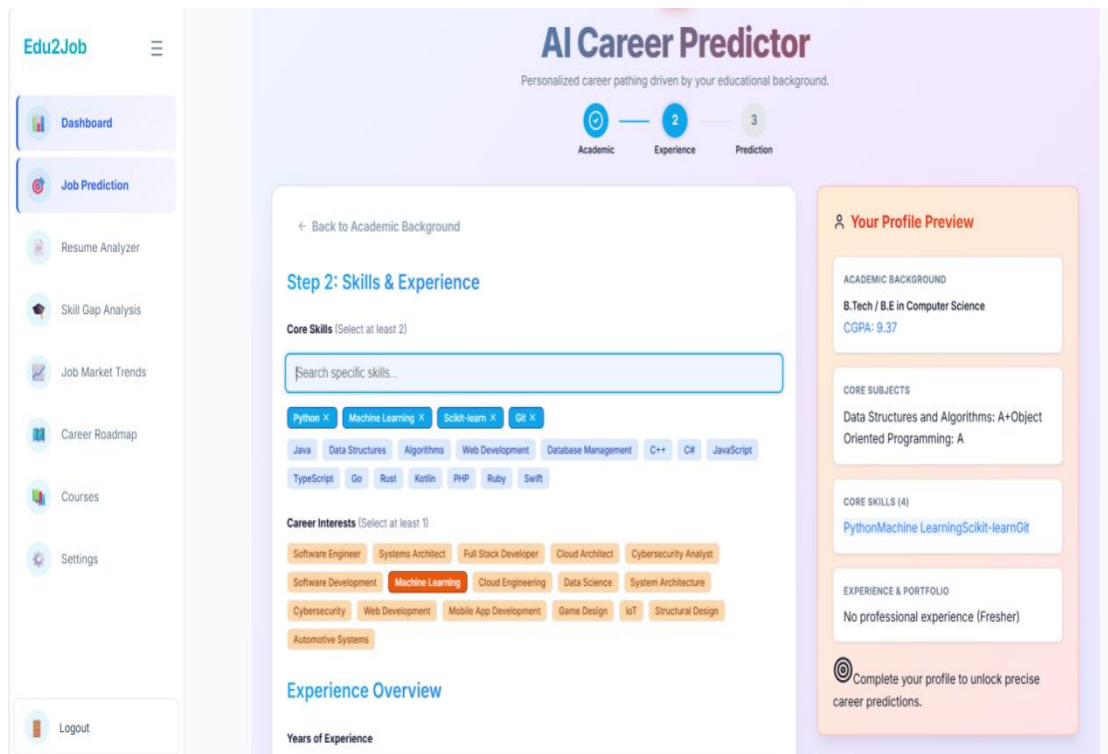


Fig. 5.4: AI Job Prediction (Skills and Experience Section)

5.4 Prediction Results Section

The Prediction Results section is the final stage of the AI Job Prediction module and represents one of the most important features of the Edu2Job – AI Career Intelligence System. After collecting academic information, technical skills, career interests, project details, internship experience, and other profile-related data, the system processes the information through the trained Gradient Boosting Classifier Machine Learning model. The model analyzes the user's complete profile and identifies career roles that best match their qualifications, skills, and interests.

The prediction engine evaluates multiple factors including educational performance, subject knowledge, programming skills, technical competencies, practical experience, and career preferences. Based on this analysis, the system generates the Top Three Career Matches and ranks them according to their compatibility with the user's profile. The role with the highest compatibility score is displayed as the Best Match, while the remaining two roles are presented as alternative career options. This ranking system helps users explore different career opportunities that align with their capabilities and interests.

The prediction results generated by the system serve as the foundation for other intelligent modules such as Skill Gap Analysis, Career Roadmap Generation, Course Recommendations, and AI Career Mentoring. These modules utilize the prediction output to provide personalized guidance and structured career development plans.

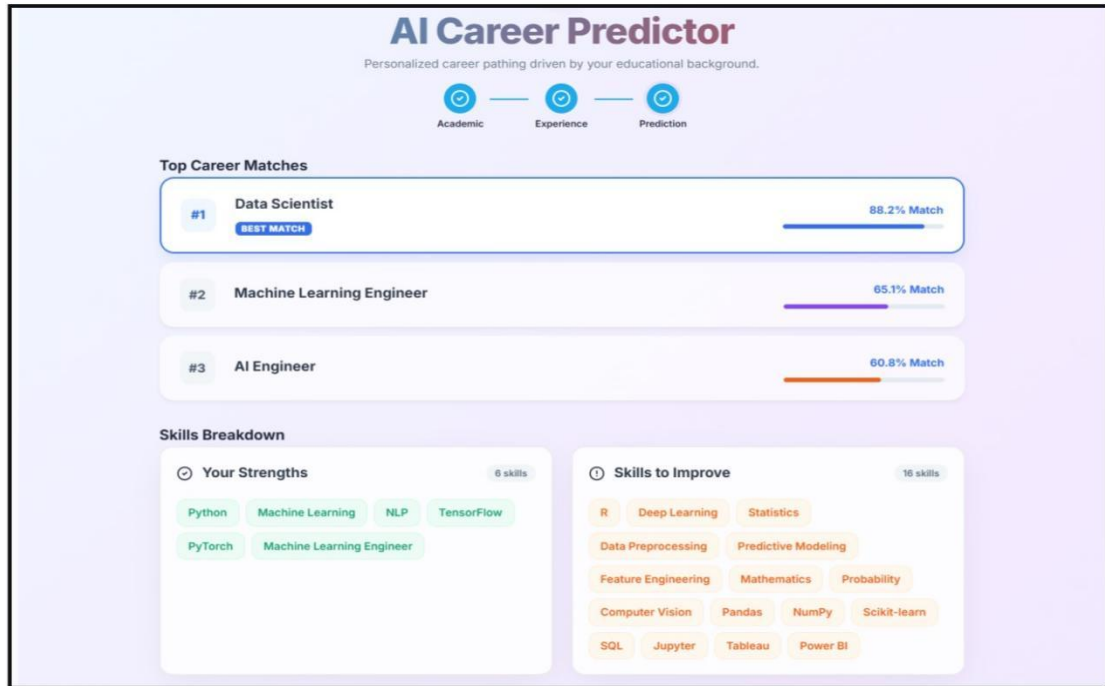


Fig. 5.5: Career Prediction Results

Overall, the Prediction Results section successfully converts educational and professional information into meaningful career insights. By leveraging Machine Learning techniques and intelligent profile analysis, the system helps students identify suitable career paths, understand industry expectations, and take informed steps toward achieving their professional goals..

5.5 Resume Analyzer Module

The Resume Analyzer Module is one of the most intelligent and feature-rich components of the Edu2Job – AI Career Intelligence System. It acts as a bridge between a student's uploaded resume and the platform's Machine Learning-based career prediction engine. The module combines Natural Language Processing (NLP), Machine Learning, and Generative Artificial Intelligence to automatically analyze resumes, identify strengths and weaknesses, predict suitable career roles, and provide personalized recommendations for profile improvement.

The analysis process begins when a user uploads a resume in PDF or text format. The system first validates the uploaded file and extracts its textual content using automated document processing techniques. For PDF files, text is extracted and cleaned before further processing. The extracted content is then passed to an NLP-based feature extraction engine that identifies important profile information such as educational qualifications, technical skills, work experience, certifications, projects, and achievements.

The backend processing layer utilizes a customized NLP pipeline developed using SpaCy. The system employs phrase matching and pattern recognition techniques to detect more than one hundred technical skills spanning domains such as software development, machine learning, cloud computing, cybersecurity, databases, frontend and backend tools. Educational qualifications, years of experience, certifications, and project details are also identified through intelligent parsing mechanisms.

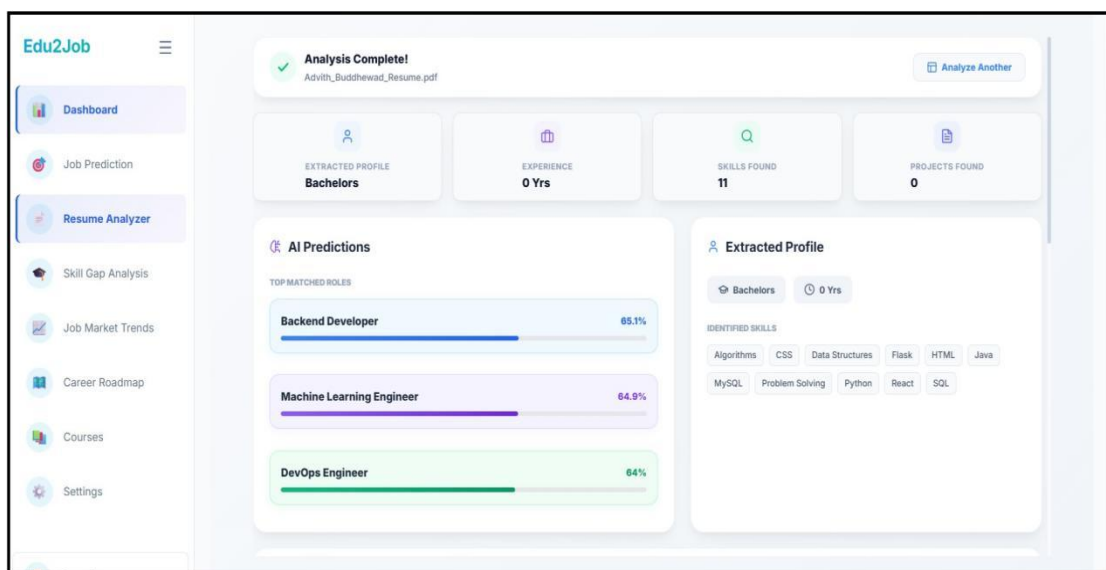


Fig. 5.6: Resume Analyzer and AI-Powered Career Dashboard.

After profile extraction, the Resume Analyzer generates a comprehensive dashboard displaying key metrics related to the user's profile. The dashboard presents information such as extracted education level, years of experience, number of identified skills, and project count. These metrics provide users with a quick overview of the information successfully detected from their resume. The extracted profile section additionally displays all recognized technical skills, allowing users to verify

the accuracy of the analysis and understand how their profile is being interpreted by the system.

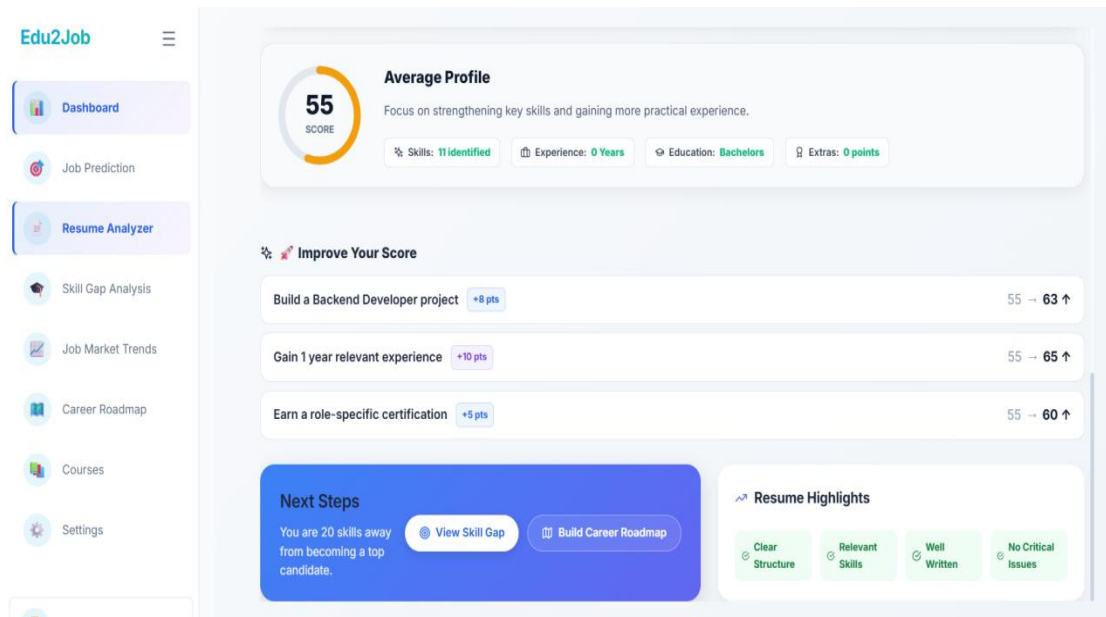


Fig. 5.7: Resume Analyzer score

The module also performs a detailed Skill Gap Analysis by comparing the user's identified skills with the expected requirements of the predicted job roles. Missing competencies are categorized into essential technical skills and additional tools or technologies that are commonly required in the industry.

Overall, the Resume Analyzer Module transforms a traditional resume review process into an intelligent career assessment system. By combining Natural Language Processing, Machine Learning, predictive analytics, and Generative AI, the module not only evaluates resumes but also provides actionable recommendations, career predictions, skill-gap identification, profile scoring, and personalized growth strategies. This makes it one of the most valuable features of the Edu2Job platform and a powerful tool for improving student employability and career readiness.

5.6 Skill Gap Analysis Module

The Skill Gap Analysis module is designed to identify the difference between a user's existing capabilities and the requirements of their target career role. This functionality helps students understand what skills they currently possess and which competencies must be acquired to achieve their desired career goals.

The system first analyzes the user's profile and predicted career role. It then compares existing skills against industry-standard skill requirements collected from career datasets and job market trends. The comparison process identifies strengths, weaknesses, and missing technologies.

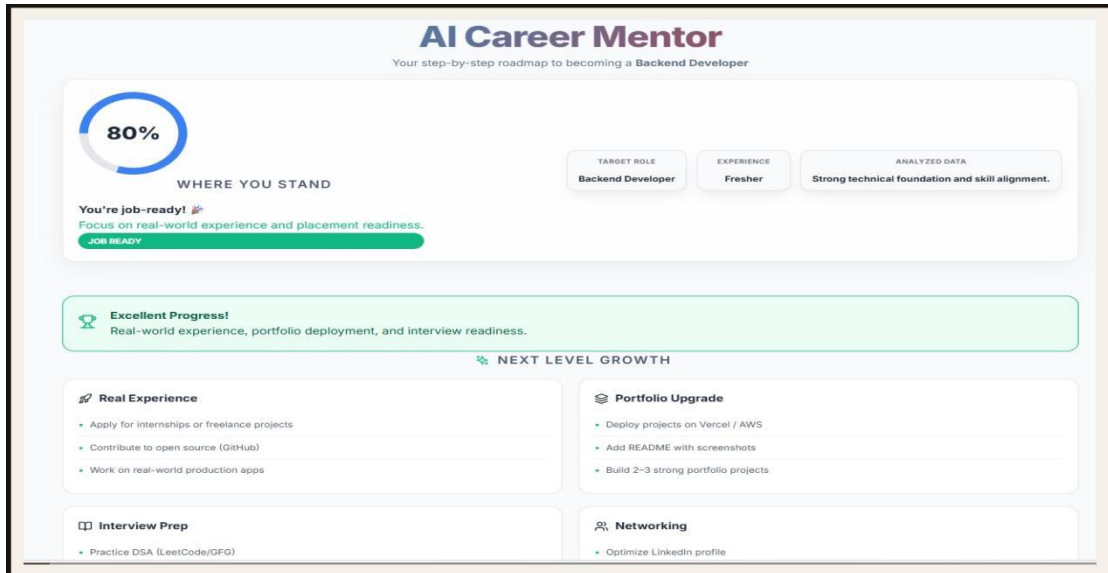


Fig. 5.8: Skill Gap Analysis

The module generates detailed recommendations for closing identified skill gaps. Missing technical skills, certifications, tools, frameworks, and industry knowledge areas are presented in an organized manner. Users can clearly understand the learning path required to become job-ready.

This module plays a crucial role in bridging the gap between education and industry expectations by helping students create focused and effective skill development plans..

5.7 AI Career Roadmap Module

The The AI Career Mentor module provides personalized career guidance based on the user's profile, prediction results, and skill readiness level. This feature acts as a virtual career advisor that continuously guides users toward their target job role. The roadmap generation process analyzes educational qualifications, technical skills, projects, certifications, and experience before creating a structured development plan. The generated roadmap includes recommendations related to practical projects, internships, portfolio development, certifications, networking activities, and interview preparation.

The system also evaluates career readiness and presents a progress indicator showing how close the user is to achieving job readiness. Different sections focus on real-world experience, portfolio enhancement, interview preparation strategies, and professional networking opportunities. This module transforms career planning into a structured and measurable process, helping students achieve career objectives through systematic progression and continuous improvement.

The report includes target role information, timeline estimates, skill requirements, recommended tools, project suggestions, certifications, and learning resources. Each stage of the roadmap is organized according to a specific learning period, allowing users to track progress effectively. The generated strategy emphasizes practical learning through project development, portfolio creation, certification completion, and interview preparation activities. Users receive a clear understanding of what actions must be performed during each stage of their career development journey.

5.8 Recommended Courses Module

The Recommended Courses module provides curated learning resources tailored to the user's career objectives and identified skill gaps. The recommendation engine analyzes career predictions, skill assessments, and roadmap requirements before suggesting appropriate courses.

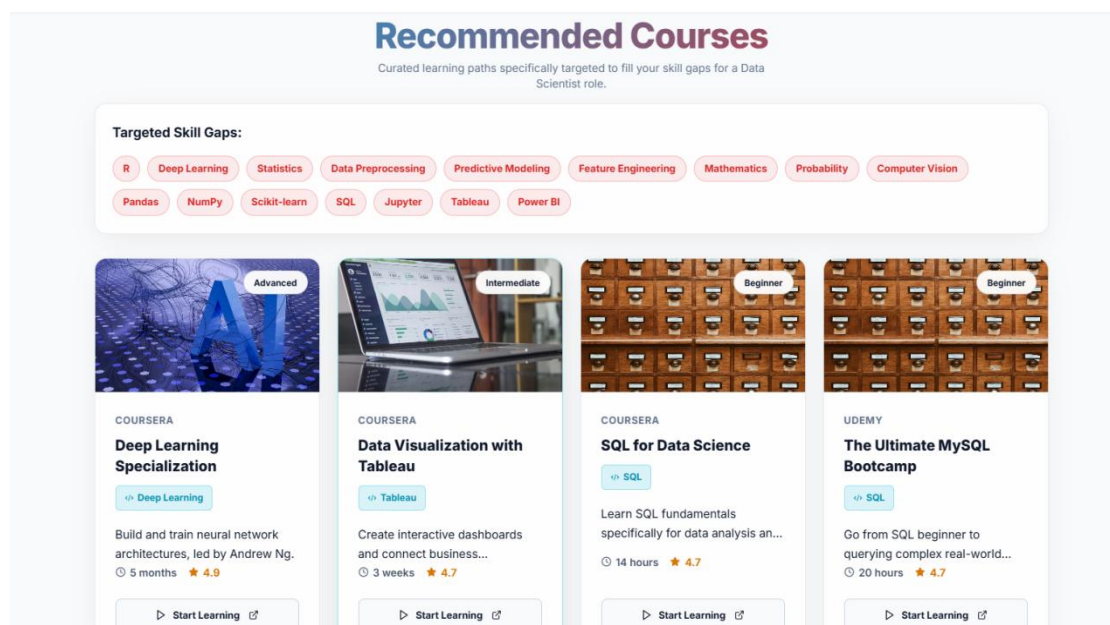


Fig. 5.9: Career Roadmap

Each course recommendation includes important information such as course title, provider, difficulty level, estimated duration, ratings, and technology category. The displayed courses cover various domains including programming languages, web development, cloud computing, data science, Machine Learning, DevOps, and software engineering. The recommendation system ensures that users focus on learning resources directly relevant to their target career roles. This eliminates unnecessary learning and helps students invest time in acquiring skills that improve employability and career readiness.

By integrating personalized course recommendations with career prediction and skill-gap analysis, the Edu2Job platform provides a complete ecosystem for continuous learning and professional growth.

5.9 Admin Dashboard Module

The Admin Dashboard Module serves as the central control and monitoring system of the Edu2Job platform. It is designed exclusively for administrators and provides a comprehensive overview of platform activities, user engagement, career prediction statistics, and system performance. Through advanced analytics and visualization tools, the dashboard transforms raw user and prediction data into meaningful insights that help administrators monitor platform growth and make data-driven decisions. The module is protected through role-based authentication and can only be accessed by authorized administrators.

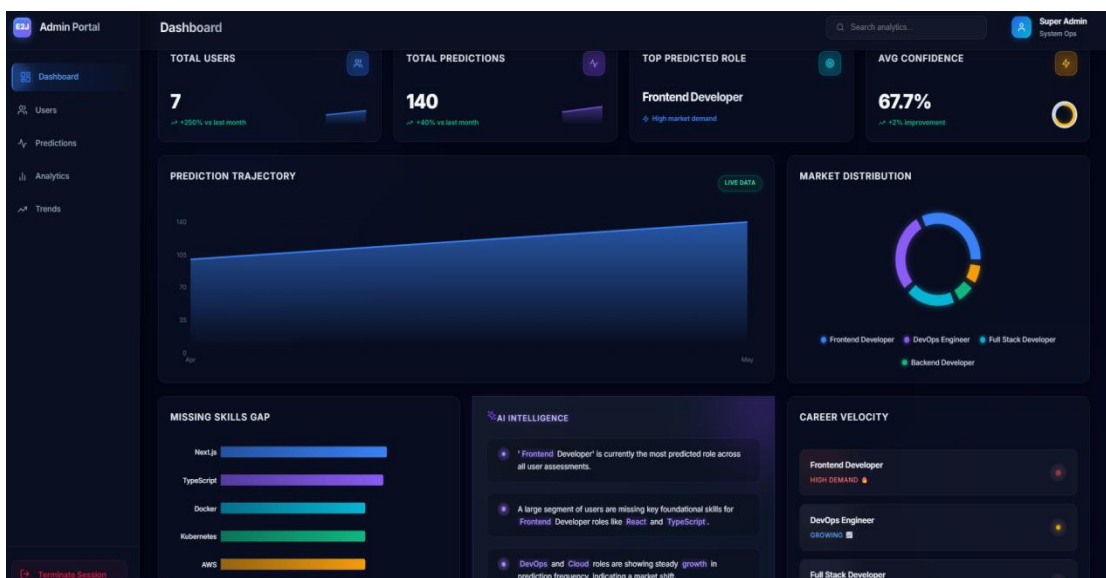


Fig. 5.10:Admin Dashboard

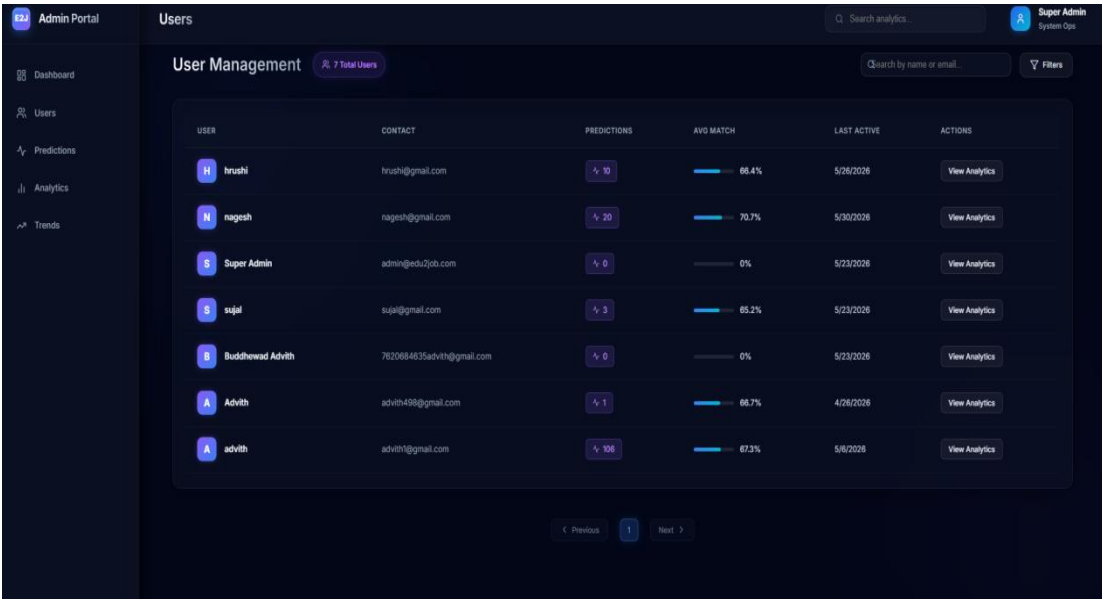
The dashboard displays various key performance indicators including total registered users, total career predictions generated, average prediction confidence, and the most frequently predicted career role. These statistics are presented through interactive cards, charts, and graphs that provide a quick summary of the platform's overall performance. Administrators can instantly understand user activity trends and the effectiveness of the AI prediction system.

Overall, the Admin Dashboard acts as an intelligent analytics hub that provides complete visibility into user activities, prediction outcomes, and career intelligence data, making it an essential component of the Edu2Job platform.

5.10 Registered Users Management Section

The Registered Users Management Section enables administrators to monitor and manage all users registered on the Edu2Job platform. This section provides a detailed view of each user's profile information, including name, email address, registration date, prediction history, average prediction confidence score, and recent activity. The information is fetched directly from the MongoDB database and displayed through a structured and searchable user management interface.

Administrators can search, filter, and paginate user records to quickly locate specific users. The system also calculates useful metrics such as total predictions generated by each user, average prediction accuracy, and last active date.



The screenshot shows the 'Users' management interface in the Admin Portal. It features a sidebar with navigation links for Dashboard, Users, Predictions, Analytics, and Trends. The main content area is titled 'User Management' and displays a table of 7 total users. The table includes columns for User (with profile icons), Contact (email), Predictions (count and status), Avg Match (percentage and progress bar), Last Active (date), and Actions (a 'View Analytics' button for each user). The users listed are hrushi, nagesh, Super Admin, sajal, Budheshwar Advith, Advith, and advith.

USER	CONTACT	PREDICTIONS	AVG MATCH	LAST ACTIVE	ACTIONS
hrushi	hrushi@gmail.com	10	66.4%	5/26/2026	View Analytics
nagesh	nagesh@gmail.com	20	70.7%	5/30/2026	View Analytics
Super Admin	admin@edu2job.com	0	0%	5/23/2026	View Analytics
sajal	sajal@gmail.com	3	65.2%	5/23/2026	View Analytics
Budheshwar Advith	7620694635advith@gmail.com	0	0%	5/23/2026	View Analytics
Advith	advith498@gmail.com	1	66.7%	4/26/2026	View Analytics
advith	advith1@gmail.com	106	67.3%	5/16/2026	View Analytics

Fig. 5.11:Registered Users

A unique feature of this section is the User Analytics Deep-Dive Page, where administrators can access detailed information about individual users. This includes prediction history, career readiness score, strengths, missing skills, roadmap progress, and AI-generated summaries. This allows administrators to identify highly active users, evaluate engagement levels, and monitor user participation across the platform.

5.11 Prediction Trends and Analytics Section

One of the primary features of this section is the Role Distribution Analysis, which shows how prediction results are distributed across different job roles such as Frontend Developer, Backend Developer, Data Analyst, Data Scientist, DevOps Engineer, and AI/ML Engineer. Administrators can easily identify which career paths are most popular among students and observe changes in demand over time. The module also includes Prediction Growth Analytics, which tracks the number of career predictions generated over different months. This feature helps administrators monitor platform adoption, user engagement, and prediction system usage.

Overall, the Prediction Trends and Analytics Section transforms large volumes of prediction data into meaningful career intelligence, helping administrators understand market trends, student interests, and future employment opportunities while supporting data-driven educational planning.

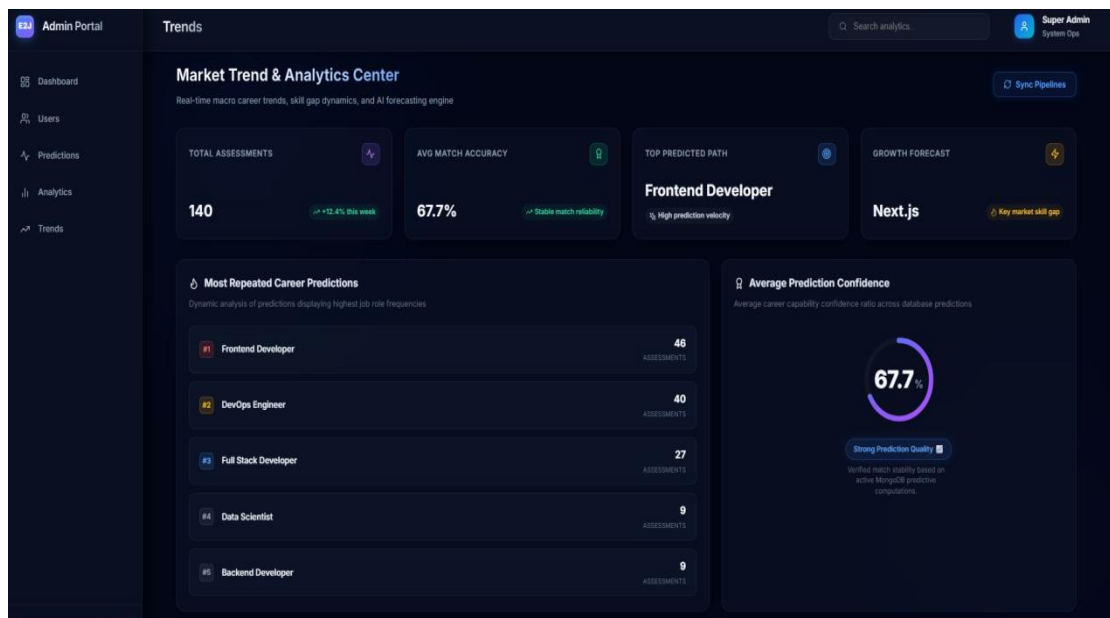


Fig. 5.12: Prediction Trends and Analytics Section

CONCLUSION

The Edu2Job – AI Career Intelligence System was developed to provide intelligent and personalized career guidance for students and job seekers. The system combines Machine Learning, Artificial Intelligence, and data analytics to predict suitable career roles based on a user's academic background, skills, certifications, projects, and professional interests.

The platform successfully integrates multiple modules, including Career Prediction, Resume Analysis, Skill Gap Analysis, Career Roadmap Generation, and Course Recommendations. Using a Gradient Boosting Classifier, the system generates accurate career predictions and helps users identify their strengths, weaknesses, and areas for improvement. The resume analysis and skill-gap modules further assist users in enhancing their profiles and improving employability.

The system was implemented using modern technologies such as React.js, FastAPI, MongoDB, Gemini AI, and Machine Learning, ensuring a scalable, secure, and user-friendly solution. Experimental results demonstrate that the platform effectively supports career planning and provides valuable insights for professional development.

In conclusion, the Edu2Job system successfully bridges the gap between academic learning and industry requirements by offering data-driven career guidance. The platform empowers users to make informed career decisions, develop relevant skills, and prepare for future employment opportunities, making it a valuable tool for career growth and professional success.

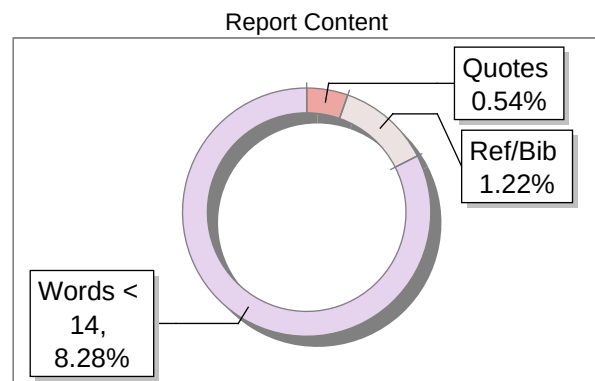
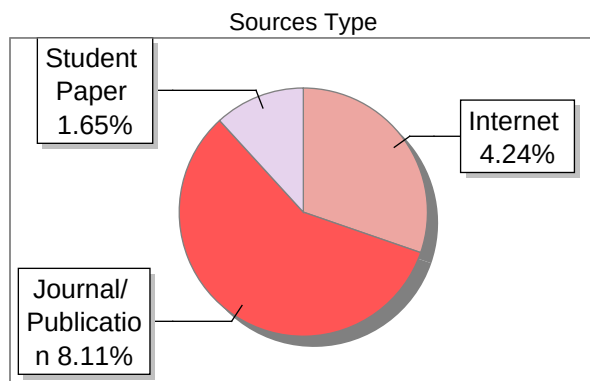
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